

Counterparty Credit Risk -- Modeling Credit Exposure

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Reference: GARP 2007 July/August Issue

A Guide to Modelling Counterparty Credit Risk

What are the steps involved in calculating credit exposure? What are the differences between counterparty and contract-level exposure? How can margin agreements be used to reduce counterparty credit risk? What is credit value adjustment and how can it be measured? **Michael Pykhtin** and **Steven Zhu** offer a blueprint for modelling credit exposure and pricing counterparty risk.

Counterparty credit risk is the risk that the counterparty to a financial contract will default prior to the expiration of the contract and will not make all the payments required by the contract. Only the contracts privately negotiated between counterparties — over-the-counter (OTC) derivatives and security financing transactions (SFT) — are subject to counterparty risk. Exchange-traded derivatives are not affected by counterparty risk, because the exchange guarantees the cash flows promised by the derivative to the counterparties.¹

Counterparty risk is similar to other forms of credit risk in that the cause of economic loss is obligor's default. There are, however, two features that set counterparty risk apart from more traditional forms of credit risk: the uncertainty of exposure and bilateral nature of credit risk. (Canabarro and Duffie [2003] provide an excellent introduction to the subject.)

In this article, we will focus on two main issues: modelling credit exposure and pricing counterparty risk. In the part devoted to credit exposure, we will define credit exposure at contract and counterparty levels, introduce netting and margin agreements as risk management tools for reducing counterparty-level exposure and present a framework for modelling credit exposure. In the part devoted to pricing, we will define credit value adjustment (CVA) as the price of counterparty credit risk and discuss approaches to its calculation.

Contract-Level Exposure

If a counterparty in a derivative contract defaults, the bank must close out its position with the defaulting counterparty. To determine the loss arising from the counterparty's default, it is convenient to assume that the bank enters into a similar contract with another counterparty in order to maintain its market position.² Since the bank's market position is unchanged after replacing the contract, the loss is determined by the contract's replacement cost at the time of default.



Contents

- ▶ **Nature of of Credit Exposure**
- ▶ **Exposure Modeling Framework**
- ▶ **Price Factor Evolution Models**
- ▶ **Valuation Models for Exposure**
- ▶ **Examples of Credit Exposure Calculation**
- ▶ **Monte-Carlo Process for Exposure Aggregation**
- ▶ **Margins and Collaterals**
- ▶ **Wrong-way and Right-way Credit Exposure**

What are risks inherent in Trading book?

▶ **Market Risk in Trading Book**

- P&L as result of adverse market moves
- VaR as a measure to monitor and control Market risk

▶ **Credit Risk in Trading Book**

- Issuer Risk, such as in bond held in book
- Settlement Risk, such as in FX spot/forward
- Pre-settlement Risk, such as in FX forward/IR Swap (CCR)

▶ **Counterparty Credit Risk (CCR)**

- It is the risk that one or more parties in a financial transaction will fail to fulfill their side of the contractual agreement, as a result of Counterparty default or its ability to meet the contractual obligations.
- CCR arises primarily from 2 main types of derivatives transactions:
 - OTC derivatives such as FX forward, interest rate swaps and swaptions.
 - SFT (eg, Securities financial transactions) such as Repo/Reverse Repo where securities are used to borrow cash or vice versa.

What is Credit Risk

Credit Risk = Exposure × Credit Worthiness × Severity

- ▶ **Exposure**

- Potential loss as a result of counterparty default.

- ▶ **Credit Worthiness**

- Probability of Default
- Risk Ratings (Moody's and S&P)

- ▶ **Severity**

- Fractional loss given default
- Severity = 1 – Recovery Rate

Types of Credit Risks

➤ **Credit Risk = EAD × PD × LGD**

TYPE OF CREDIT RISK

CAUSE OF ECONOMIC LOSS

Lending Risk	▶ Borrower Defaults ▶ Loan value decreases as a result of – Decrease in credit quality of borrower – Increase in general credit spread
Issuer Risk (Specific Risk)	▶ Issuer of security defaults ▶ Security's market value decreases – Decrease in credit quality of borrower
Counterparty Risk • Settlement • Pre-settlement	▶ Counterparty Defaults – In an exchange, we pay but don't receive – Counterparty defaults before settlement and contract has positive value to us

Market and Credit Risks

Attribute	Market Risk	Credit Risk
Time Horizon	▶ Short-Term risk (1-10 days)	▶ Long-Term Risk
Nature of Loss	▶ Adverse Market Move	▶ Loss is contingent on counterparty default when we have trading gains
Risk Mitigation	▶ Matching Positions Risk can be eliminated by offsetting positions	▶ Use of Netting Agreements ▶ Collateral ▶ Mutual Terminations ▶ Counterparty Diversification
Legal Issues	▶ Loss is not dependent on legal issues	▶ Loss magnitude is affected by enforceability of netting agreements

Transactions carry Counterparty Risk

► Pre-Settlement Risk

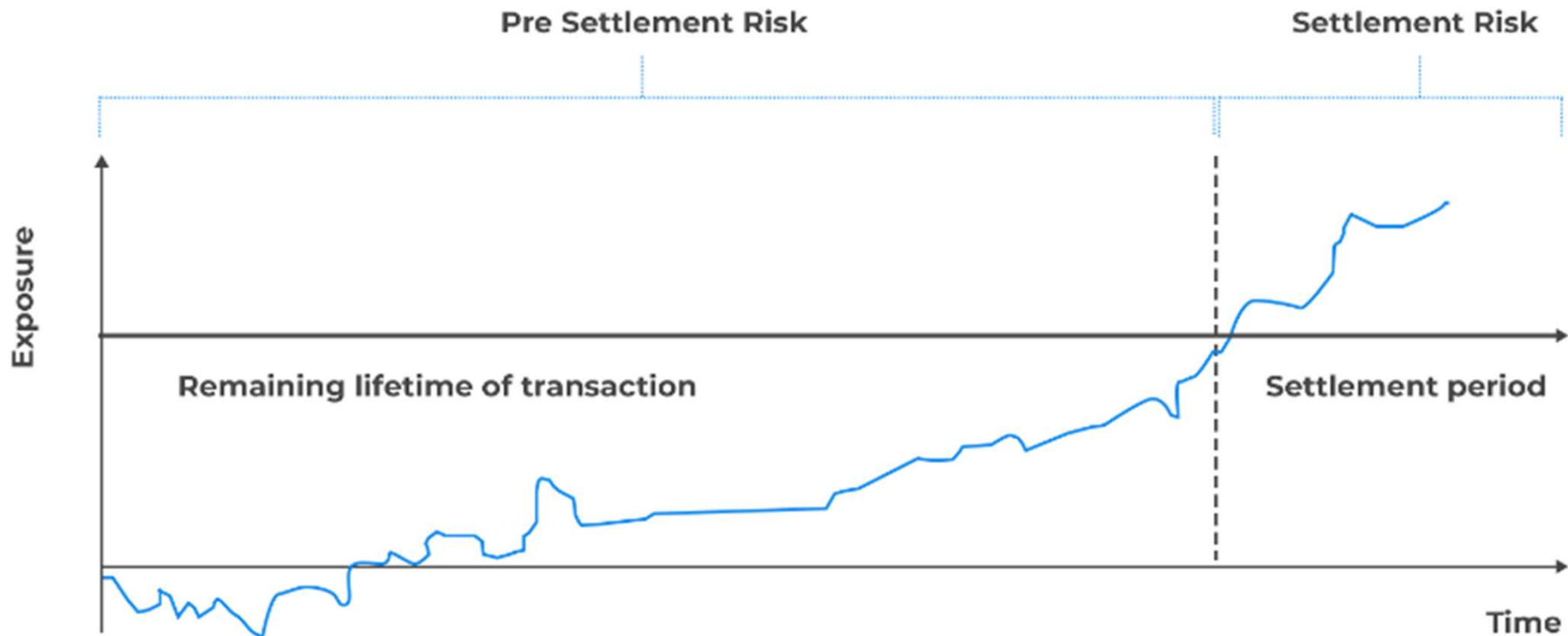
- The risk that a Counterparty will default prior to the final settlement of transaction

► Settlement Risk

- The risk that a Counterparty fails to settle (pay or deliver) according to the contractual term of transaction



Pre-settlement Risk vs. Settlement Risk



Institutions with Significant Counterparty Risk

▶ Large Derivatives Players

- Large banks have high volume of derivatives on their trading books. They often trade with each other but also have a large number of clients coming from other industry sectors.
- These banks participate in almost every derivative product available on the market, including **interest rate swaps**, **foreign exchange forwards**, and **credit derivatives**. These banks require collateral posting from Counterparties against their positions

▶ Medium Derivatives Players

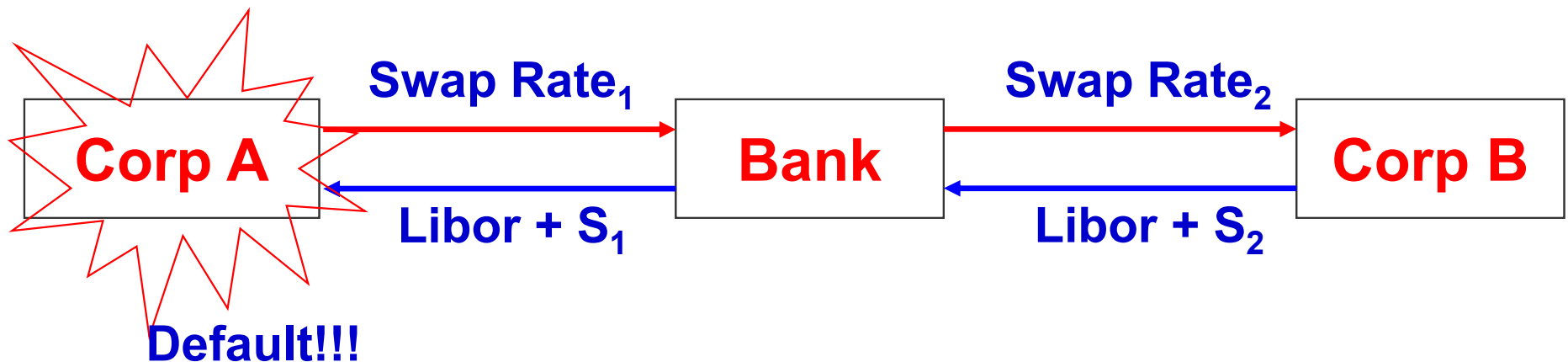
- Small-to-Medium size banks, hedge funds, and pension funds. Players in this category trade with a relatively large number of clients. They are active in derivatives market and may not always willing to post collateral against positions.

▶ Small Derivatives Players

- Smaller financial institutions, large corporates, or sovereign entities with significant derivatives requirements. For example, they may be looking to hedge their investments or reduce the borrowing cost. They usually have a limited volume of derivatives trades on their books and have a relatively small number of counterparties. They may also specialize in a certain asset class, such as foreign exchange products and interest rate swaps. They often do not have collateral operation to post collateral.

Nature of Counterparty Credit Risk

- ▶ By-product of bilateral derivatives transactions
- ▶ Create a chain of dependencies among derivatives counterparties
- ▶ Lead to credit exposures that must be measured and managed in practice



Interest Rate Swap Exposure



► Bank's Exposure

- Counterparty A defaults, LIBOR goes up and the swap has positive MTM
- Since the Bank receives Libor+Spread, the replacement value of the swap increases

Repo and Reverse Repo (抵押回购)

► How CCR arises in Repo transactions

For cash lender (buyer of securities),

- **Risk**: The borrower defaults and fails to buy back the securities at the end of repo term.
- **Credit Exposure**: The lender must then sell the security to recover the cash lent. In a market downturn, the lender faces a loss if market value of the security has fallen below the loan amount since the deal was initiated. This exposure is a function of market risk, operational risk, and the counterparty's creditworthiness.

For cash borrower (seller of securities),

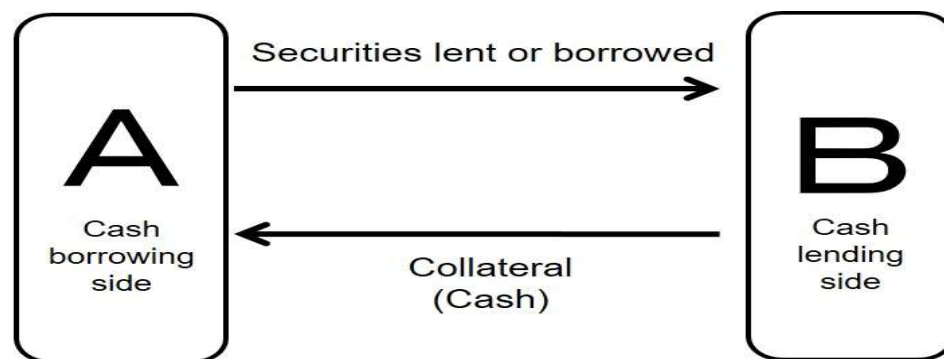
- **Risk**: The lender defaults and does not sell the securities back to the borrower as agreed.
- **Credit Exposure**: The borrower loses the securities that were posted as collateral. If the value of the collateral is greater than the cash received, the borrower experiences a loss.

► Repo transactions (short-term secured loan)

Repo transaction is executed on GRMA, which is equivalent to ISDA for OTC derivatives.

- **Repo haircut** is a **Risk Management** tool to protect the cash lender in event of borrower's default
- **Repo rate** is interest rate paid on the cash borrowed, set as difference btw initial sale price and repurchase price of the security

Central bank (Fed) use the repo rate as a monetary policy tool to influence the money supply and short-term interest rates.



FX Forward and Interest rate Swaps (利率互换)

▶ Foreign Exchange Forward (外汇远期交易)

- Bank bears the pre-settlement risk (CCR) with exposure (PSE) equal to difference in market value btw Receive currency (eg, EUR) and Pay currency payments (eg, USD).
- $MtM(t) = \exp[-r(T - t)] \cdot (Notl = \$100m) \cdot [Fwd(t) - K]$, for $t \leq T$
- CCR could materialized anytime during the life of the contract, when counterparty defaults.

▶ Interest Rate Swaps (利率互换)

- IR swap also carry CCR because counterparty may fail to make fixed or floating interest payments as required, even though no exchange of notional principal on swap



Interest Rate Swap with a Swap Dealer



Why Interest rate Swaps carry CCR?

► Interest Rate Swaps

- For pay-fixed swap, MtM is computed as discounted cashflow on difference btw the floating-rate payments and the fixed-rate payments
- $MtM(t) = (Notl = \$100m) \cdot [\sum_{t_i \geq t} F_i(t) b_i d_i(t) - K \sum_{t_i \geq t} b_i d_i(t)]$
 $= (Notl = \$100m) \cdot D(t) \cdot [S(t) - K]$, for $D(t) = \sum_{t_i \geq t} b_i d_i(t) \sim$ Swap Duration
- For receive-fixed swap, MtM is computed as discounted cashflow on difference btw the fixed-rate payments and the floating-rate payments
- $MtM(t) = (Notl = \$100m) \cdot [K \sum_{t_i \geq t} b_i d_i(t) - \sum_{t_i \geq t} F_i(t) b_i d_i(t)]$
 $= (Notl = \$100m) \cdot D(t) \cdot [K - S(t)]$, for $D(t) = \sum_{t_i \geq t} b_i d_i(t) \sim$ Swap Duration



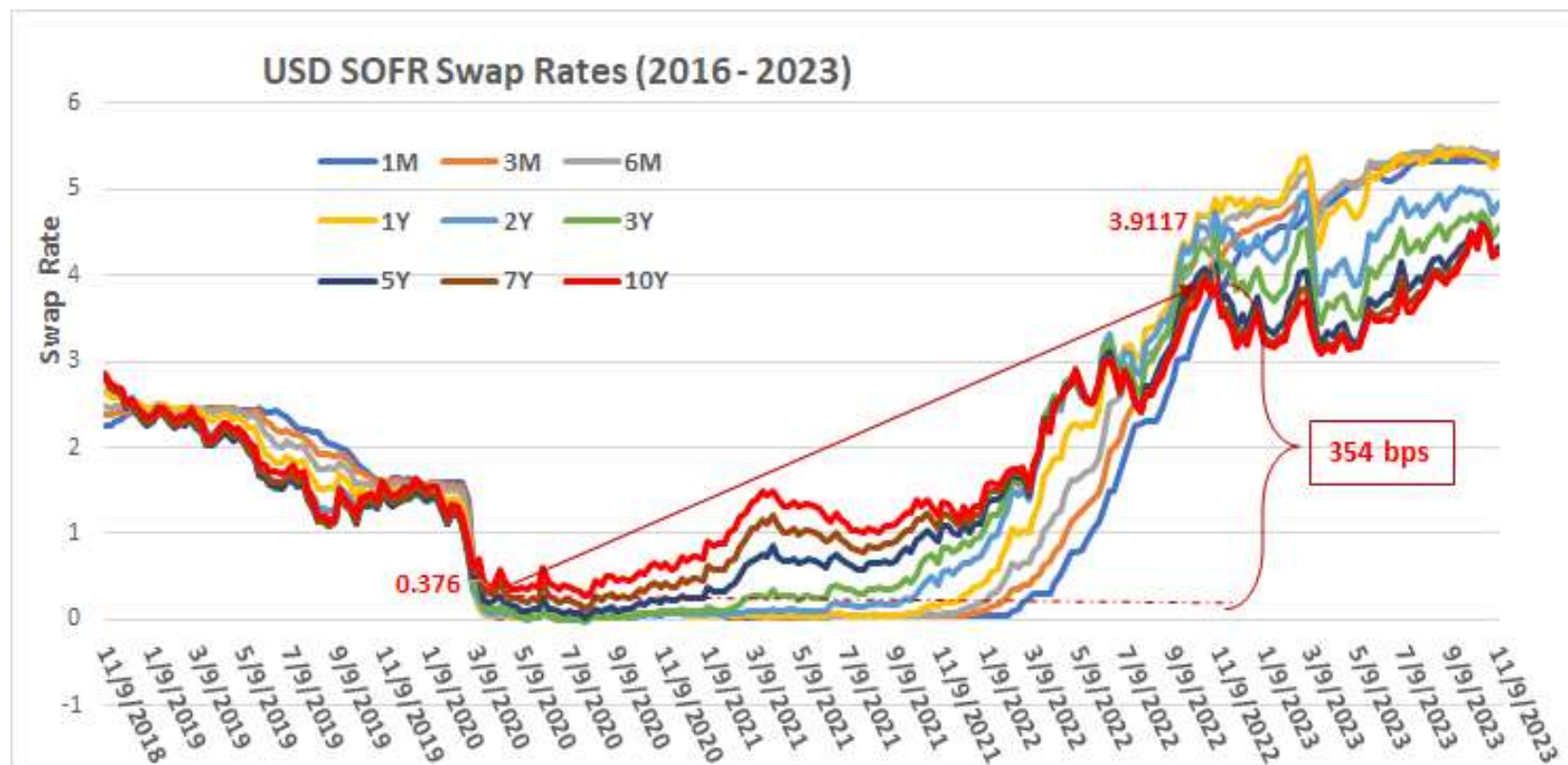
Interest Rate Swap with a Swap Dealer



How much Swap rates can move?

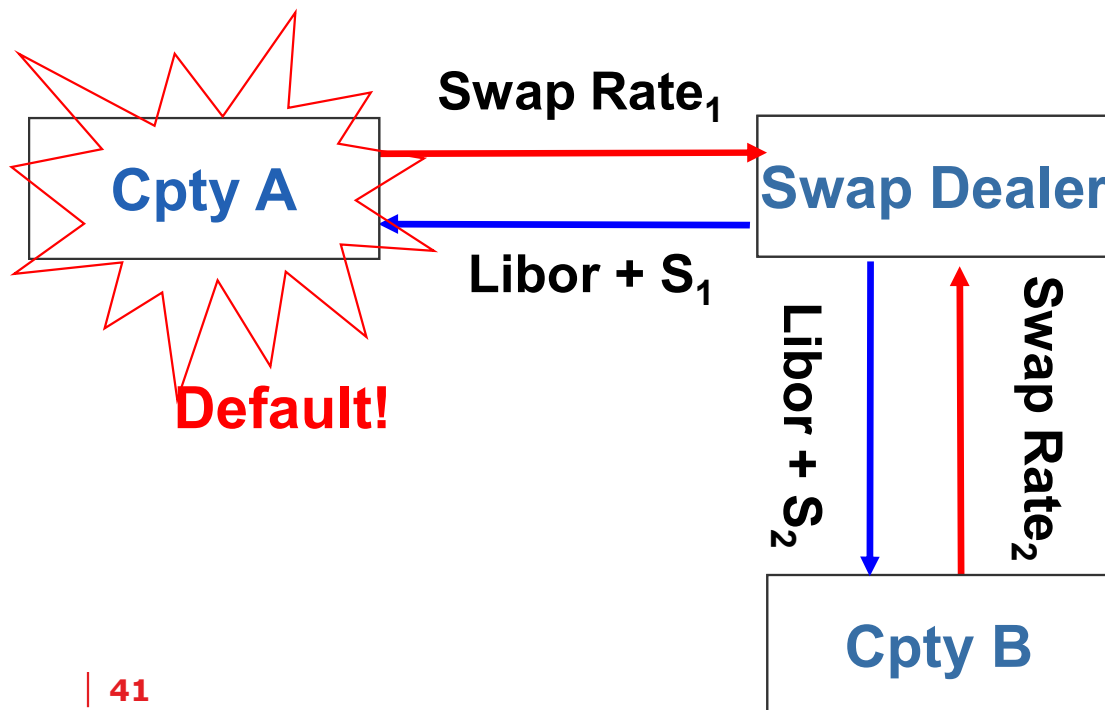
► Counterparty Credit Risk of Interest Rate Swap

- 10r Swap rate $S(t)$ increased ~354 bps during Covid period (3/2020 – 11/2022)
- For a payer swap on \$100m, swap has 6.1yrs duration for remaining life
Mark-to-Market (MtM) = \$100m x 6.1 x 3.54% = 21.6m or 21.6% of Notional.
- Dealer could face potential loss of MtM value on the swap if Counterparty defaulted

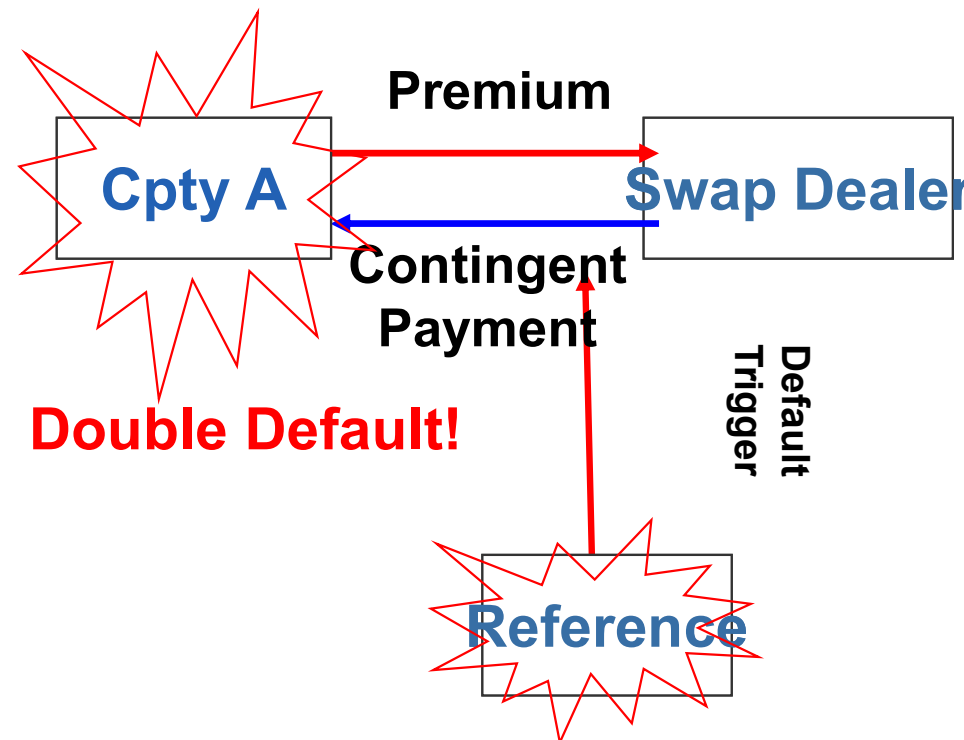


Counterparty Credit Risk - Examples

- ▶ **Interest Rate Swaps** carry credit risk because Counterparty may fail to make floating/fixed cash payments as and when required. However, risk is reduced since there is no exchange of principal and further reduced through netting/ offsetting of the fixed-floating cashflows exchanged periodically.



- ▶ **Credit Default Swaps** carry credit risk also in the form of **wrong-way risk (WWR)** when exposure to Counterparty is adversely correlated with credit quality of that Counterparty.



CCR – Wrong Way Risk (错向风险)

- ▶ **WWR (错向风险)** when exposure to a counterparty adversely correlated with credit quality of counterparty
 - It is risk when default risk and credit exposure tend to increase together (eg, Upward slope in Graph).
 - **WWR** results in an increase of CCR and hence it increases **CVA** and reduces **DVA** (eg, reduction to derivative value due to the Bank's own default risk)
- ▶ **Right-way risk (正向风险)** when credit exposure is favorably correlated with credit quality of counterparty.
 - Default risk in RWR transaction tends to decrease when credit exposure increases (eg, Downward slope in Graph)
 - **RWR** results in a decrease of CCR and hence it reduces **CVA** and increases **DVA**

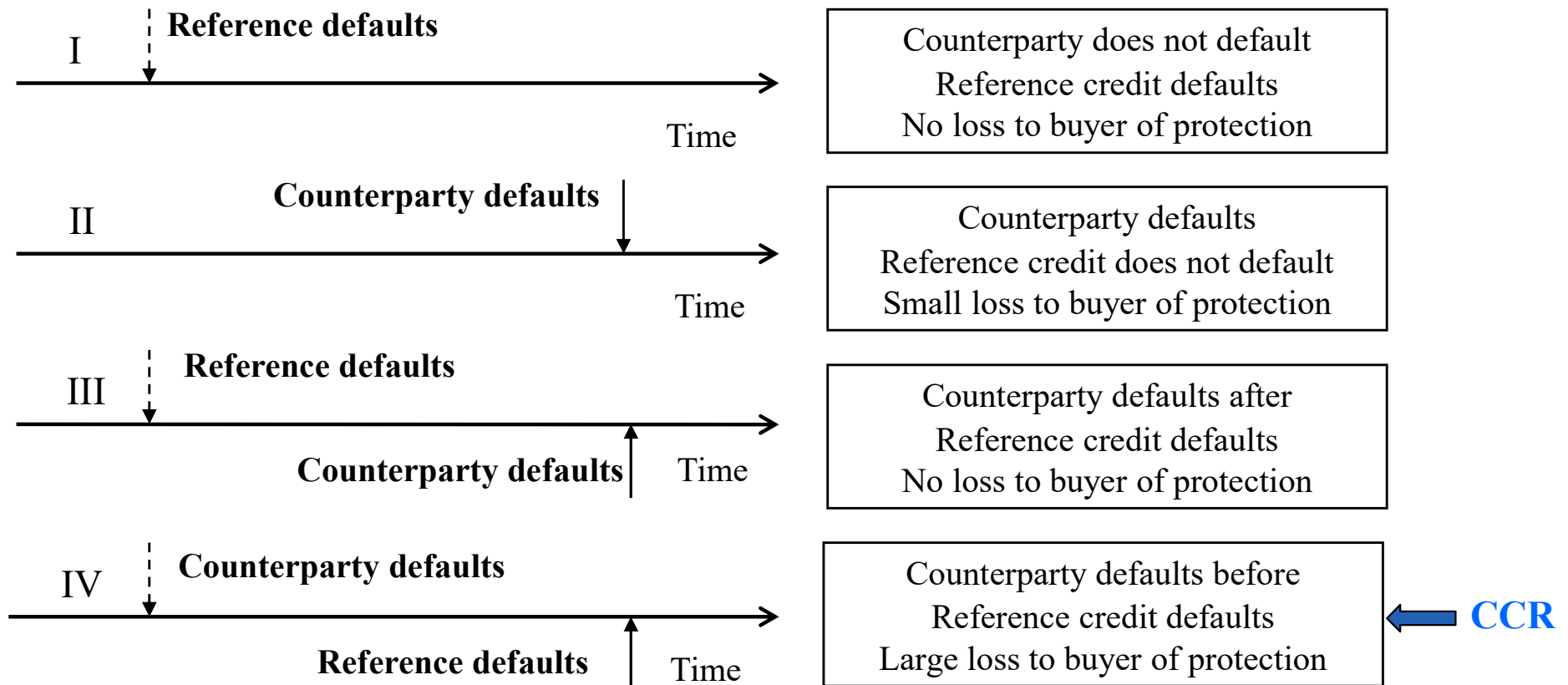


Wrong/Right-Way Risk

- ▶ Conditioning on default is important when there is a relation between changes in exposure and changes in counterparty's credit quality – ***Wrong-way*** or ***Right-way risk***
- ▶ **Wrong-way risk**: counterparty's credit quality worsens under the same market conditions that lead to higher exposure
 - Equity: we buy a put option on the stock of the counterparty.
 - Commodity: we enter into a receiver oil forward/swap with oil producer.
 - Credit: we buy protection on the counterparty.
- ▶ **Right-way risk**: counterparty's credit quality worsens under the same market conditions that lead to lower exposure
 - Equity: we buy a call option on the stock of the counterparty.
 - Commodity: we enter into a payer oil forward/swap with oil producer.
 - Credit: we sell protection on the counterparty.

Credit Default Swap (信用违约互换)

- ▶ **CDS has counterparty credit risk in the form of wrong-way risk**
 - **ISDA** defines WWR when “exposure to a counterparty is adversely correlated with the credit quality of that counterparty.”
- ▶ **Default Scenarios between Counterparty and Reference Credit**



Institutions with Significant Counterparty Risk

▶ Large Derivatives Dealers

- Large banks have high numbers of derivatives on their books. They often trade with each other but also have a large number of clients coming from other sectors within the finance industry. These banks participate in just about every derivative product available on the market, including interest rate swaps, foreign exchange forwards, and credit derivatives. Dealers in this category almost always post collateral against their positions

▶ Medium Derivatives Players

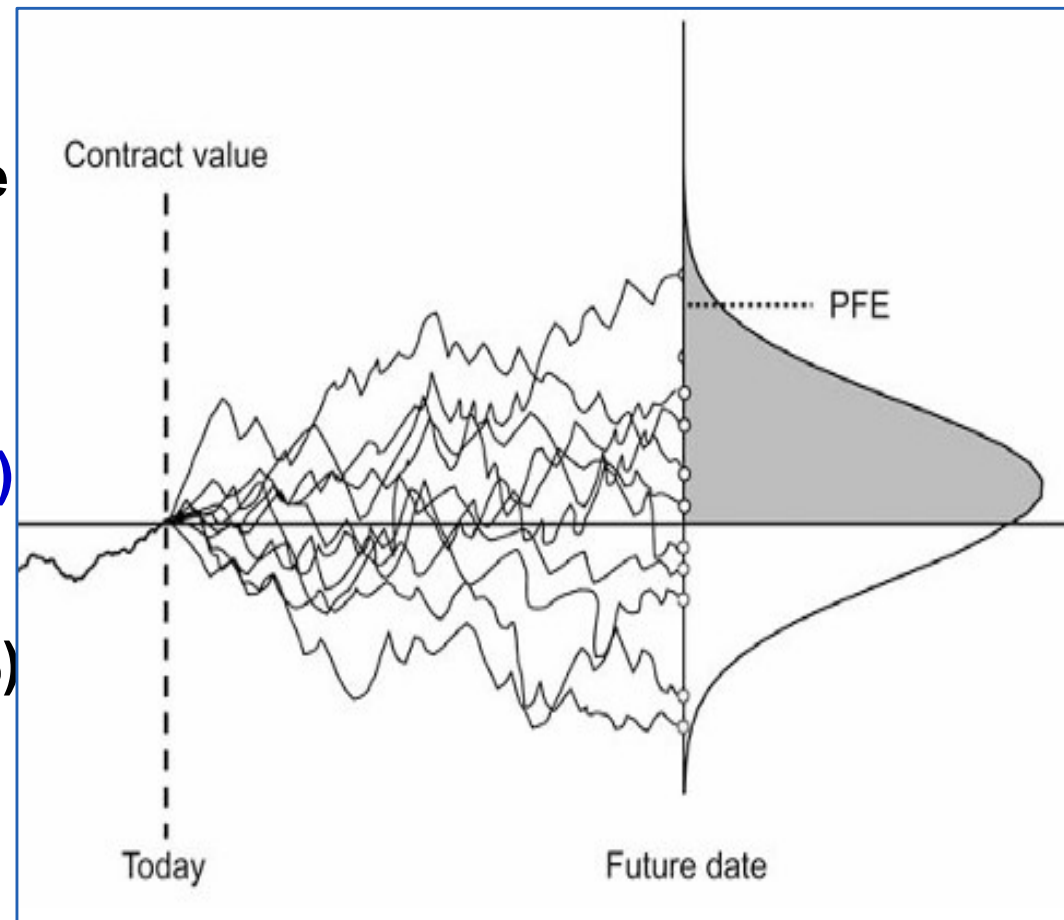
- Regional banks, hedge funds, and pension funds. Players in this category trade with their clients. They are active in a large derivatives market but they may not always post collateral against positions.

▶ Small Derivatives Participants

- Small financial institutions, large corporates, or sovereign entities with significant derivatives requirements. For example, they may be looking to hedge their loan obligation. They have just a few derivatives trades on their books and have a relatively small number of counterparties. They may also specialize in a certain asset class, foreign exchange/interest rate swaps, and often **do not post collateral**.

Credit Exposure

- ▶ Amount that we can lose over a given period as a result of counterparty default. Exposure does not consider (directly) any recovery nor the probability of default.
- ▶ Current Exposure
 - Current replacement cost (amount that we can lose if the counterparty defaults today)
 - $\text{Max}(\text{current MTM}, 0)$
- ▶ Potential Future Exposure (PFE)
 - Maximum amount of exposure on a future date at a given statistical confidence (eg, 95%)
 - Credit limits are usually set against PFE



Exposure Profile and Credit Limits

► Exposure Profiles

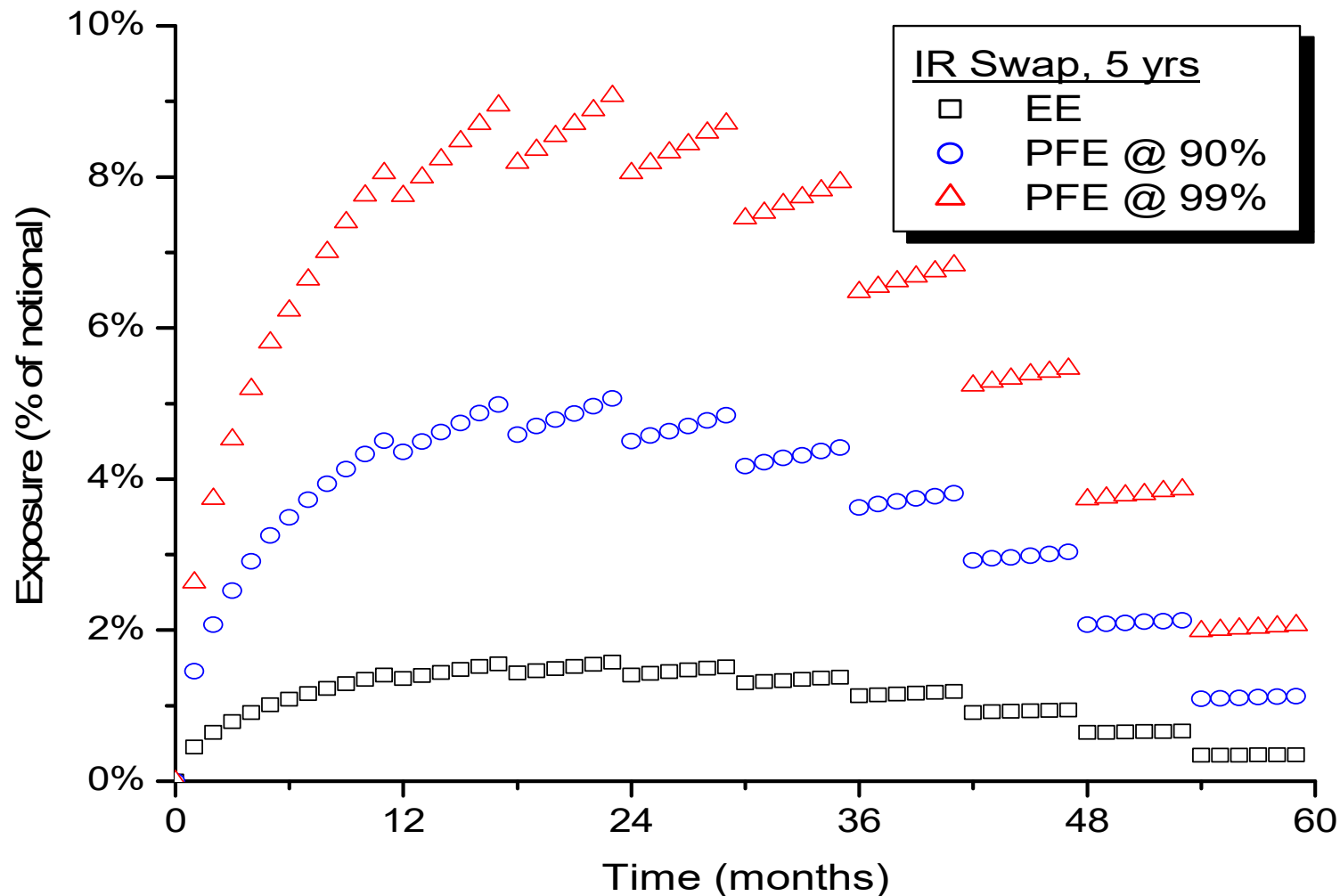
- Potential future exposure (PFE): future exposure at some high confidence level (e.g., 95%)
- Expected exposure (EE): expectation of the future exposure for all future dates
- Derivative Risk Exposure (DRE): square root of expectation of exposure squared for all future dates

► Credit Limits

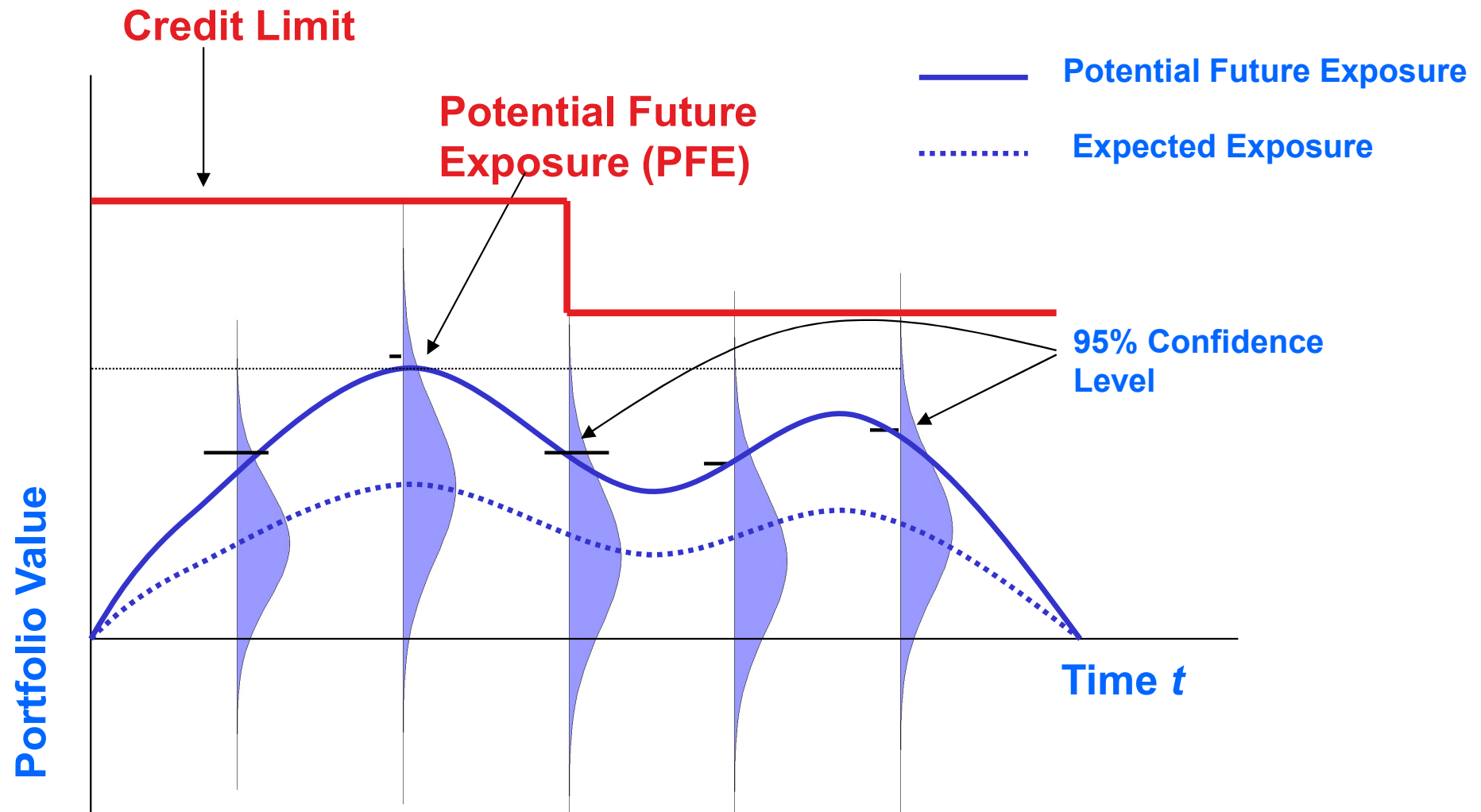
- Credit Limit is often set at each Product Level of Cpty
- Exposure Tenor and Exposure Amount are set according to Bank's internal rating for Cptys.

IR Swap Example: Exposure Profiles

► Exposure profiles for a single swap



Credit Exposure and Credit Limit



Unlike traditional single-horizon risk measures such as VAR, credit exposure needs to be defined over multiple time horizons to fully understand the impact of the time and specifics of the underlying contracts.

PFE Modeling Simulation Framework

- ▶ **Simulate future states of market risk factors**
 - Scenario model to generate future market scenarios
 - Generate correlated future scenarios of relevant risk factors (e.g., FX, Credit Spreads, Interest Rates, etc.)
- ▶ **Value each deal in the portfolio**
 - Valuation model to evaluate transaction for each scenario
 - Using simulated market risk factors as input to pricing models
- ▶ **Aggregate counterparty exposures**
 - Use appropriate netting rules, collateral, and other credit enhancements
- ▶ **Calculate the exposure measures**
 - Confidence level exposure (e.g., 95%)
 - Expected Exposure and Effective Exposure

Scenario Generation

- Price factor Evolution Models
- Case Studies: Demo in Spreadsheet

Price Factors and Risk Factors

► Price Factor

- The element of market data that is required for deal pricing
- Example: EURUSD FX Rate, 6M FWD USD Interest Rate, MSFT stock price, etc.
- The value of a Price Factor at a given simulation date is used as the market data to value a deal.

► Risk Factor

- The individual stochastic source of uncertainty driving the evolution of the Price Factor.
- The number of risk factors depends on the model that describes the Price Factor evolution. This could be an actual market variable (e.g., spot FX rate), or “abstract” factor (e.g., principal component of a yield curve).

Modeling the Price Factor Evolution

► Distribution of Equity Prices or Interest Rates

– Lognormal Distribution

- return of the price or rate has a normal distribution

– Price/Rate evolution follows a Geometric Brownian Motion

$$\Delta S = \mu S \Delta t + \sigma S z \sqrt{\Delta t}$$

or alternatively

$$S_t = S_o e^{\left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + z \sigma \sqrt{t} \right]}$$

– Mean-reversion model

$$\Delta \log(S) = \lambda [\log(\bar{S}) - \log(S)] \Delta t + \sigma z \sqrt{\Delta t}$$

Ornstein-Uhlenbeck Process

► **$X = \text{Ln}(S)$ follows an OU process**

- $dx_t = \lambda(\bar{x} - x_t)dt + \sigma dW(t)$
- λ = MR speed, w = Brownian motion

► **Solving for $x(t)$**

$$x_T = e^{-\lambda T} x_0 + (1 - e^{-\lambda T}) \bar{x} + \sigma \int_0^T e^{-\lambda(T-t)} dW(t)$$

► **Mean and Variance**

- $E[x(t + dt)|x(t)] = e^{-\lambda dt} x(t) + (1 - e^{-\lambda dt}) \bar{x}$
- $V[x(t + dt)|x(t)] = \sigma^2 / 2\lambda \cdot (1 - e^{-\lambda 2dt})$

► **Auto-correlation:**

- $\text{Corr}[x(t+dt), x(t)] = \exp[-\lambda dt]$.

OU model – parameter estimation

- ▶ **Simulate the path-wise scenario at each time step**

$$x(t + dt) = e^{-\lambda dt} x(t) + (1 - e^{-\lambda dt}) \bar{x} + \sigma \sqrt{(1 - e^{-2\lambda dt}) / (2\lambda)} \cdot z, \quad \text{for } z \sim N(0,1)$$

- ▶ **Apply least-square regression on the historical time series data**

- $X_{t+dt} = \alpha + \beta X_t + \varepsilon_t$

- ▶ **Estimation of Mean-reversion and Volatility:**

- **Mean reversion rate:** $\lambda = -\text{Ln}(\beta)$, for β = the slope of the above regression
 - **Mean reversion level:** $\bar{x} = \alpha / (1 - \beta)$, for α = the intercept of the regression
 - **Volatility:** $\sigma^2 = \sigma_\varepsilon^2 \cdot 2\lambda / (1 - e^{-2\lambda dt})$, for σ_ε = stdev of regression error (ε_t).
 - **Half-Life:** $HL = \text{Ln}(2) / \lambda$, for time taken for X to return to MR-level

Modeling the Price Factor Evolution

► Mean-Reversion and Volatility Term Structure

- Forward price has volatility term structure

$$dF(t, T) = \sigma e^{-\lambda(T-t)} F(t, T) dW_t$$

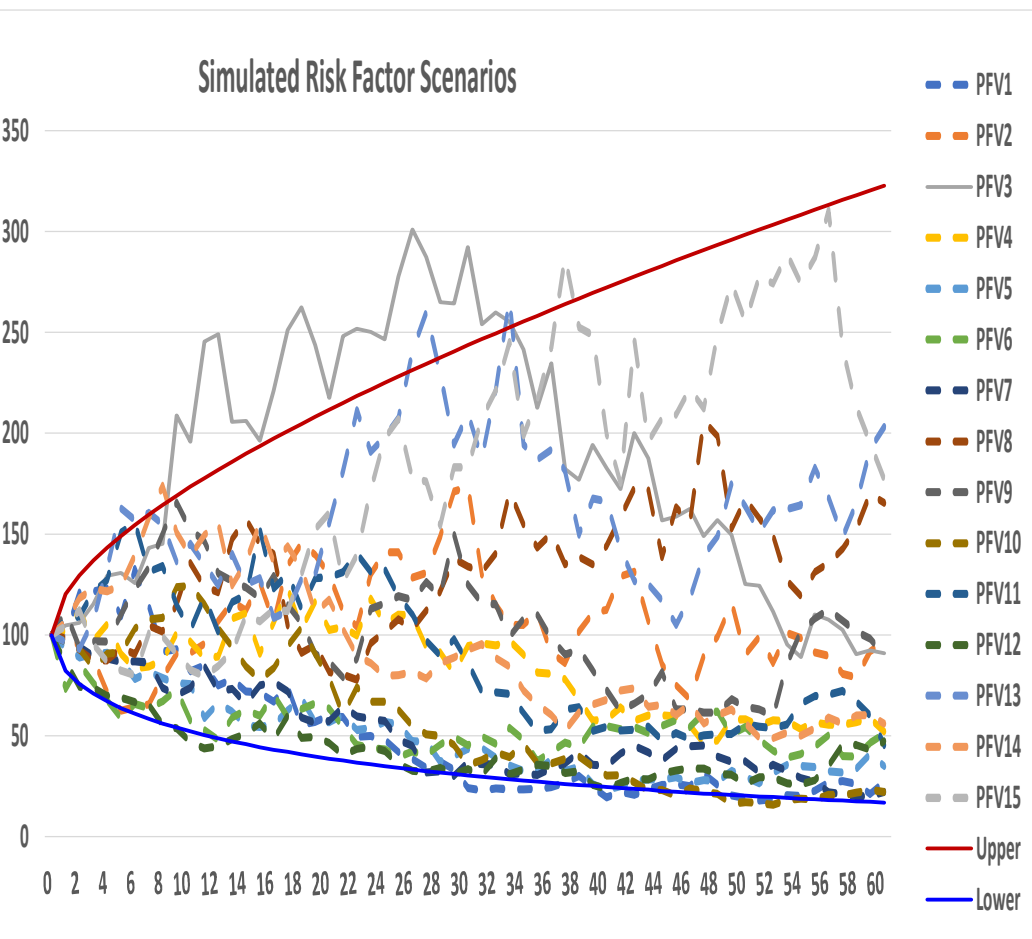
- Spot price $S(t) = F(t, t)$ is mean-reverting

$$dS(t) = \lambda[F(0, T) - S(t)]dt + \sigma S(t) dW_t$$

Modeling the Price Factor Evolution

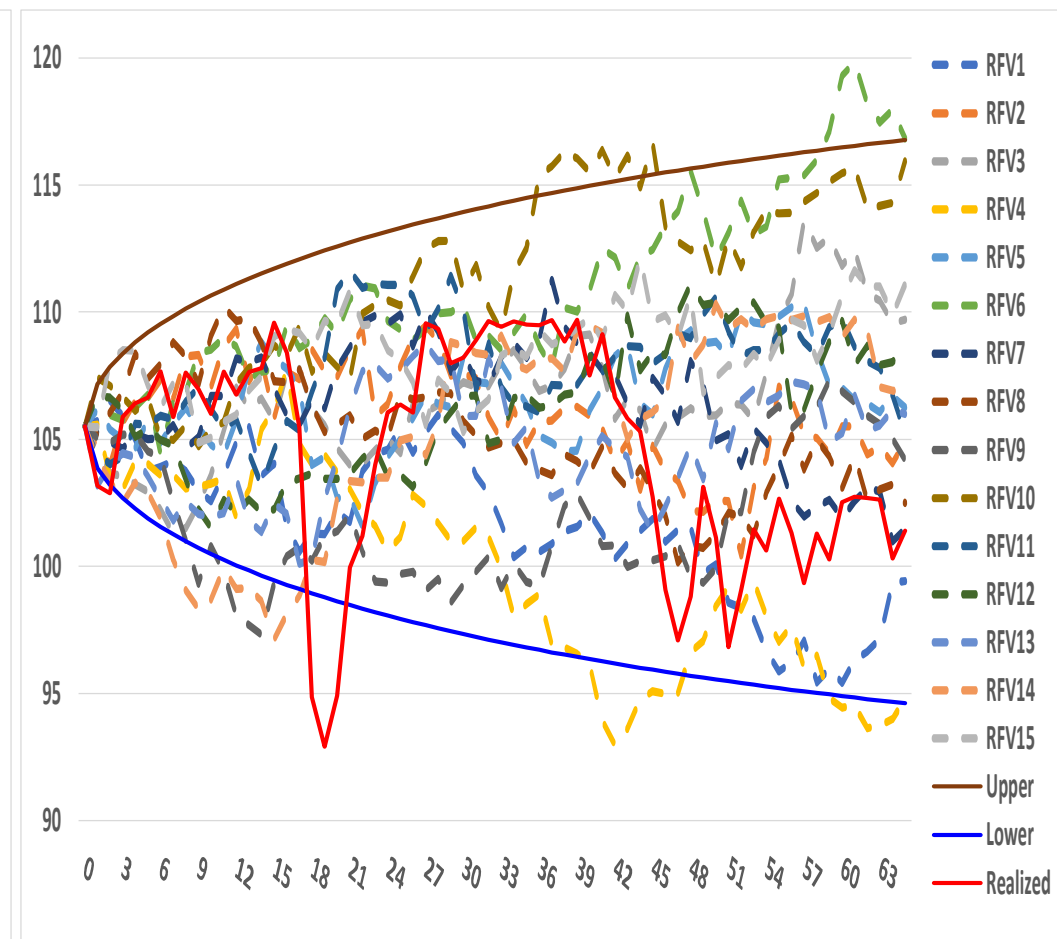
Sample Path of Lognormal(LN) process

- ✓ *Drift – determine trend.*
- ✓ *Volatility - determine width of band.*



Sample Path of Ornstein-Uhlenbeck (OU) process

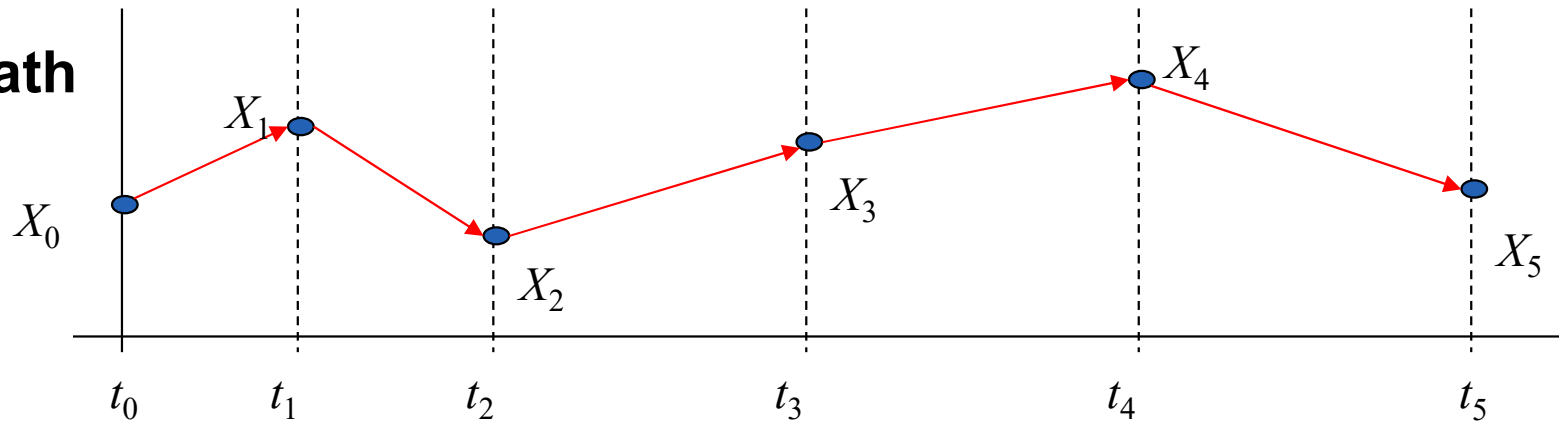
- ✓ *Mean-Reversion Rate and Level.*
- ✓ *Volatility - determine width of band.*



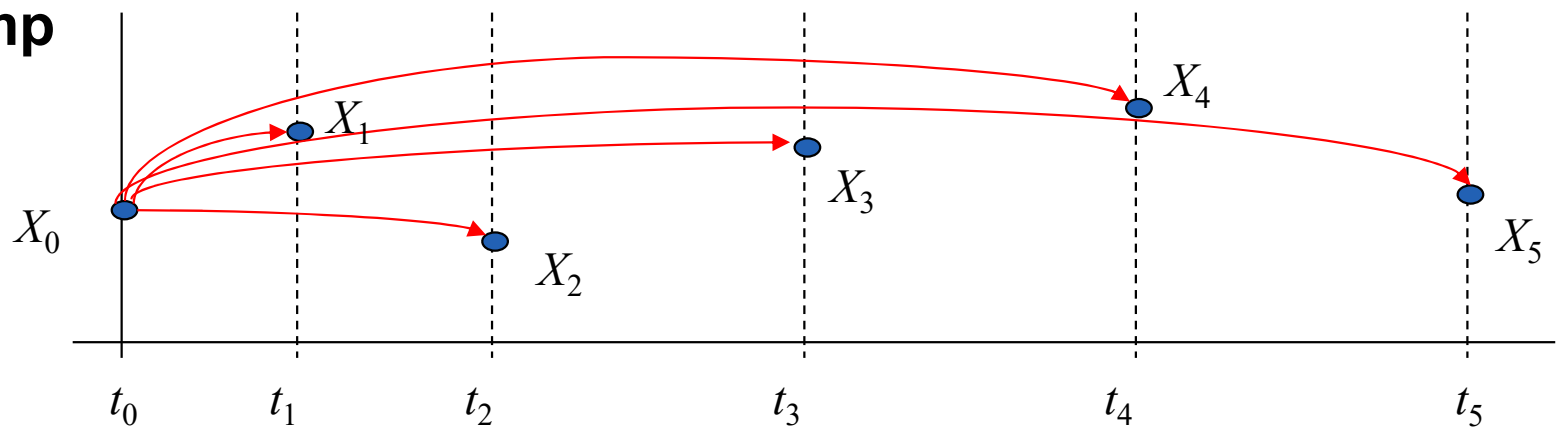
Simulating Market Scenarios

Path-by-Path vs. Direct-Jump to Simulation Date

Path-by-Path

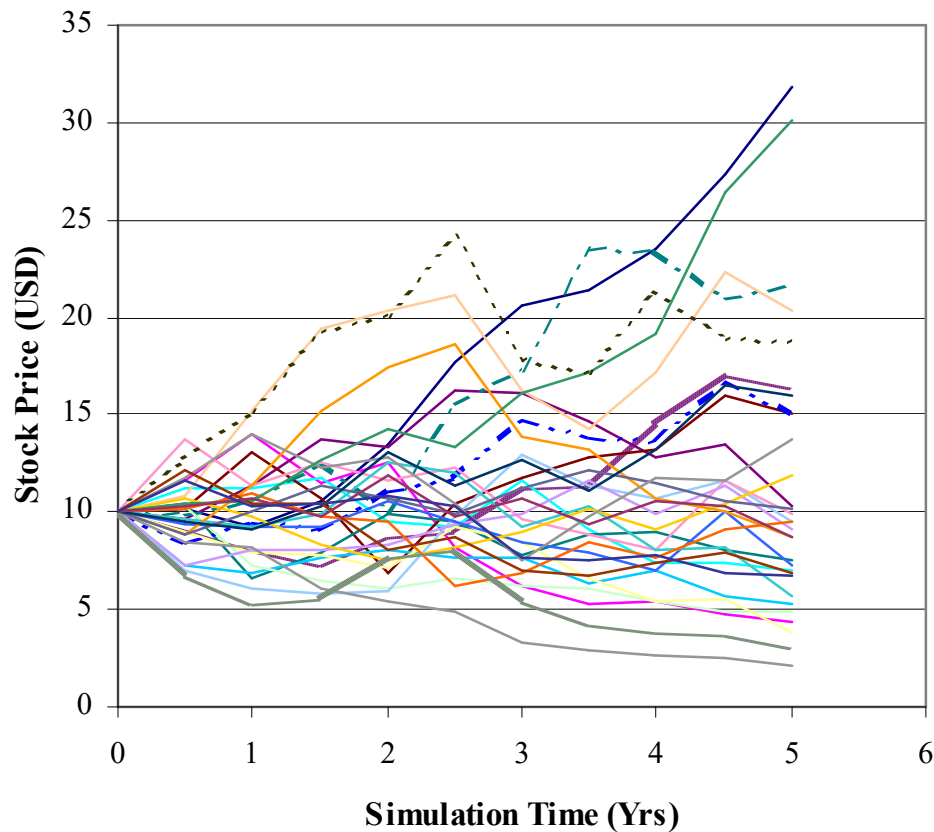


Direct Jump

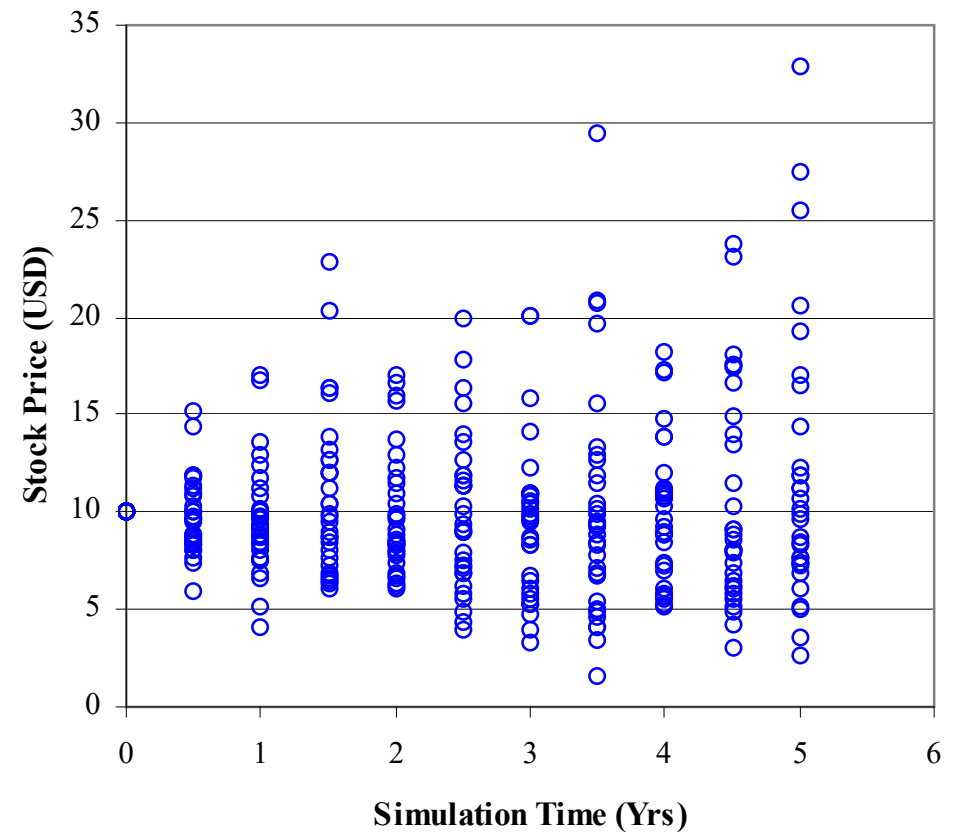


Simulating Market Scenarios

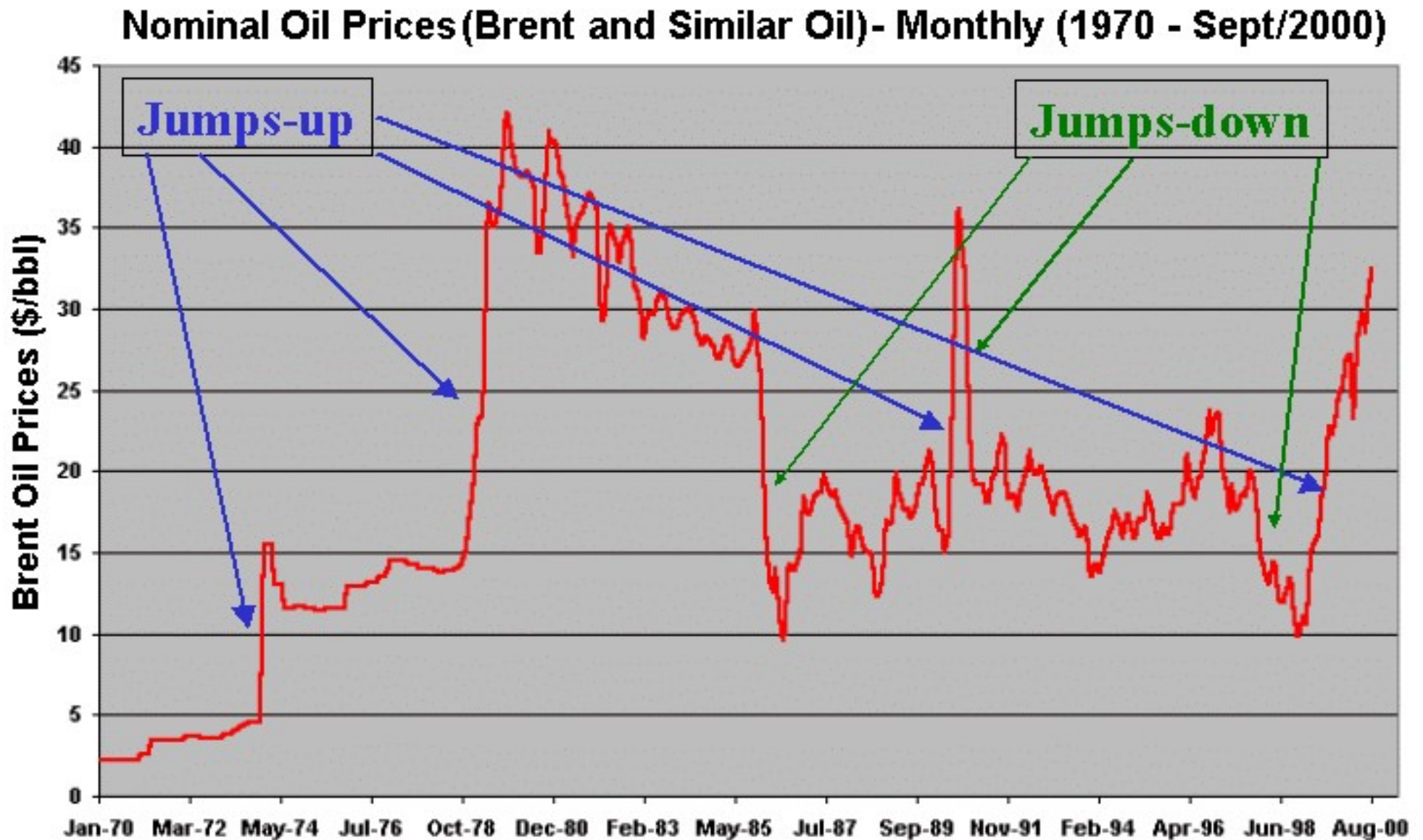
A. Path-Dependent Simulation



B. Direct-Jump to Simulation Date



How to model the jumps



Modeling the Jump-Process

► Random Jump Model

- FX in Emkt may follow a Poisson-jump process

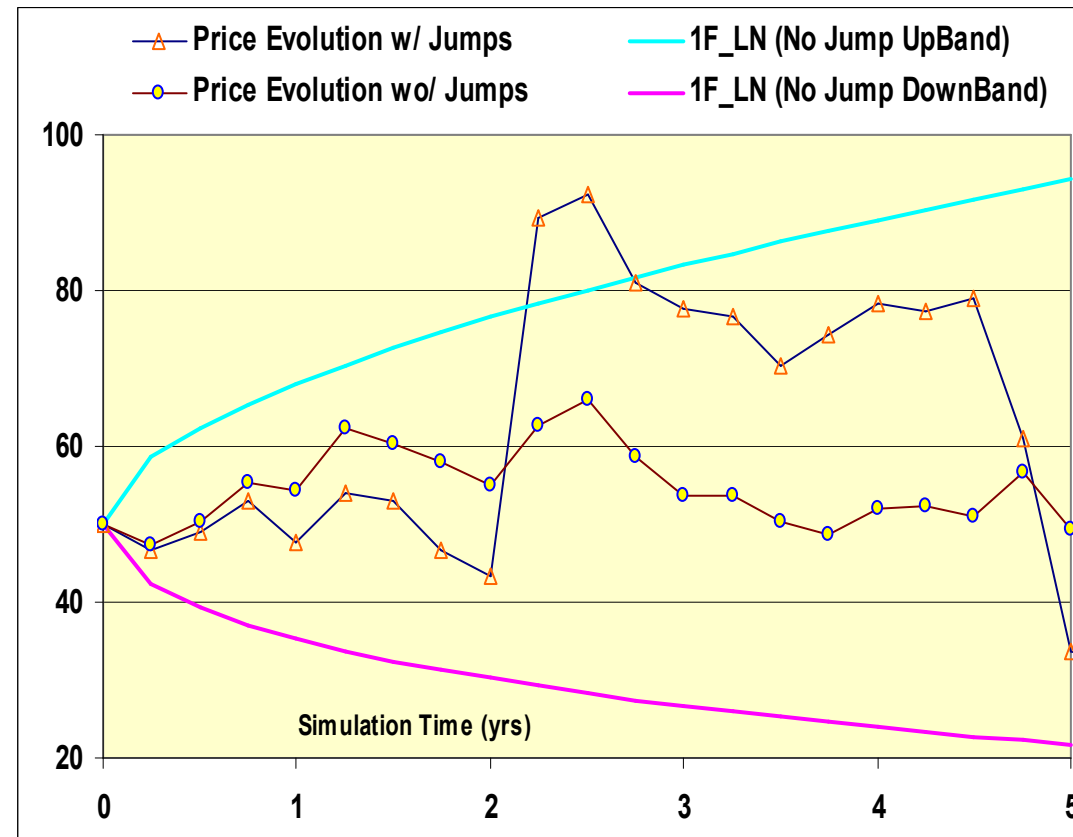
$$\Delta X_t = X_t \cdot [\mu \Delta t + (1 - \alpha) \Delta N_t]$$

or alternatively

$$X_t = X_o e^{\mu t} \alpha^{N_t}$$

- N_t is a standard Poisson process with intensity

► Jump vs No-Jump



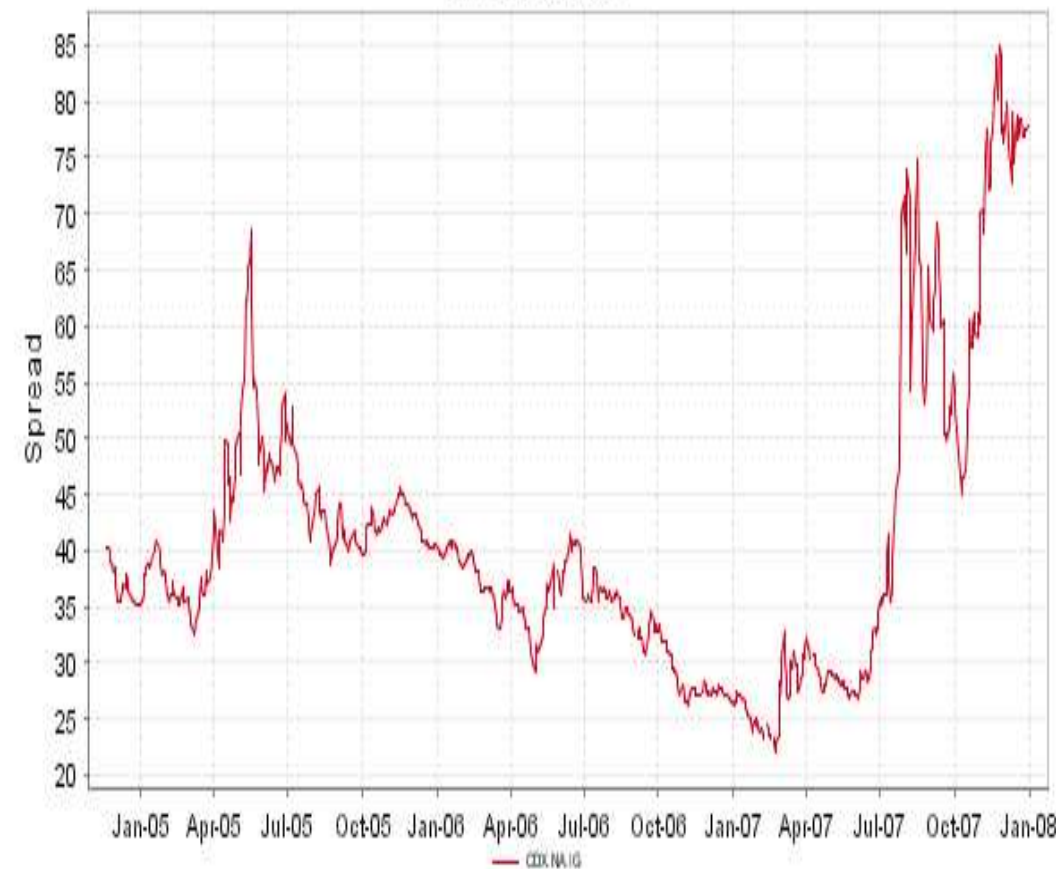
CDS Index: Historical Spreads Jan 2005 - Dec 2007

US CDX IG Spreads

-Max = 85bps

-Min = 23bps

CDS Indices



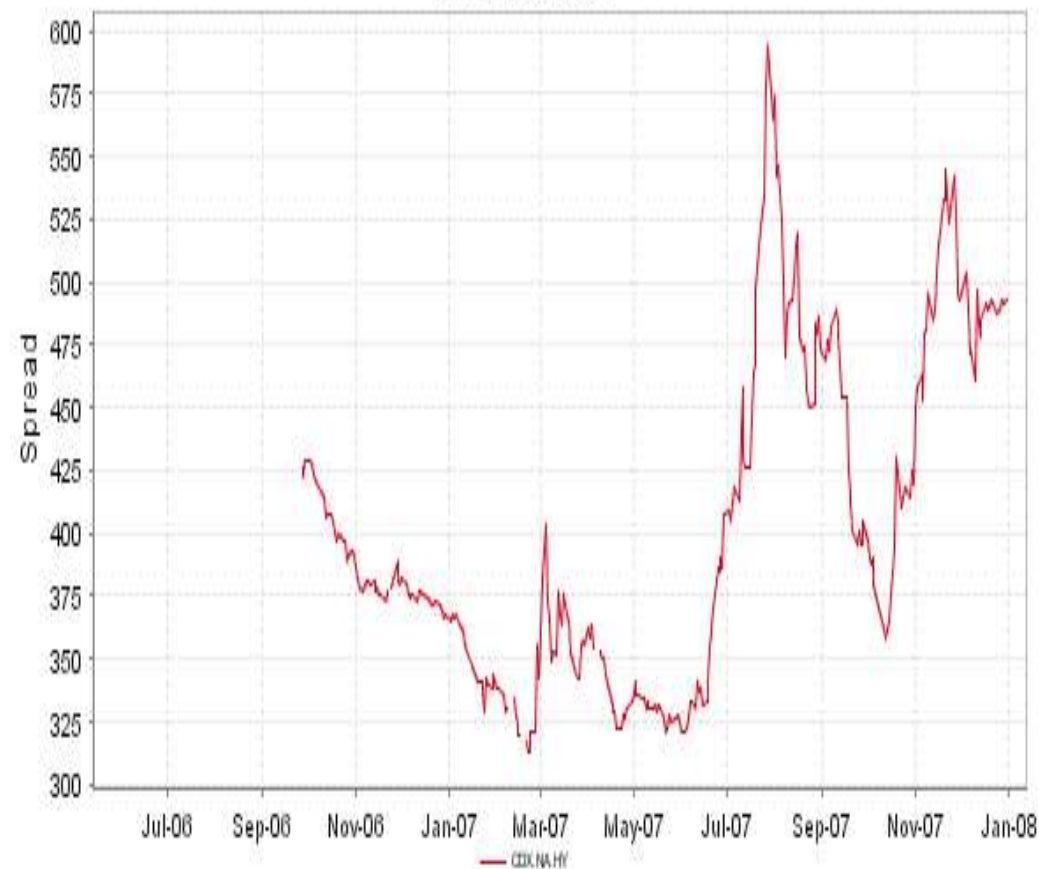
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US CDX HY Spreads

-Max = 594bps

-Min = 318bps

CDS Indices



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Historical Time Series (Credit Index)

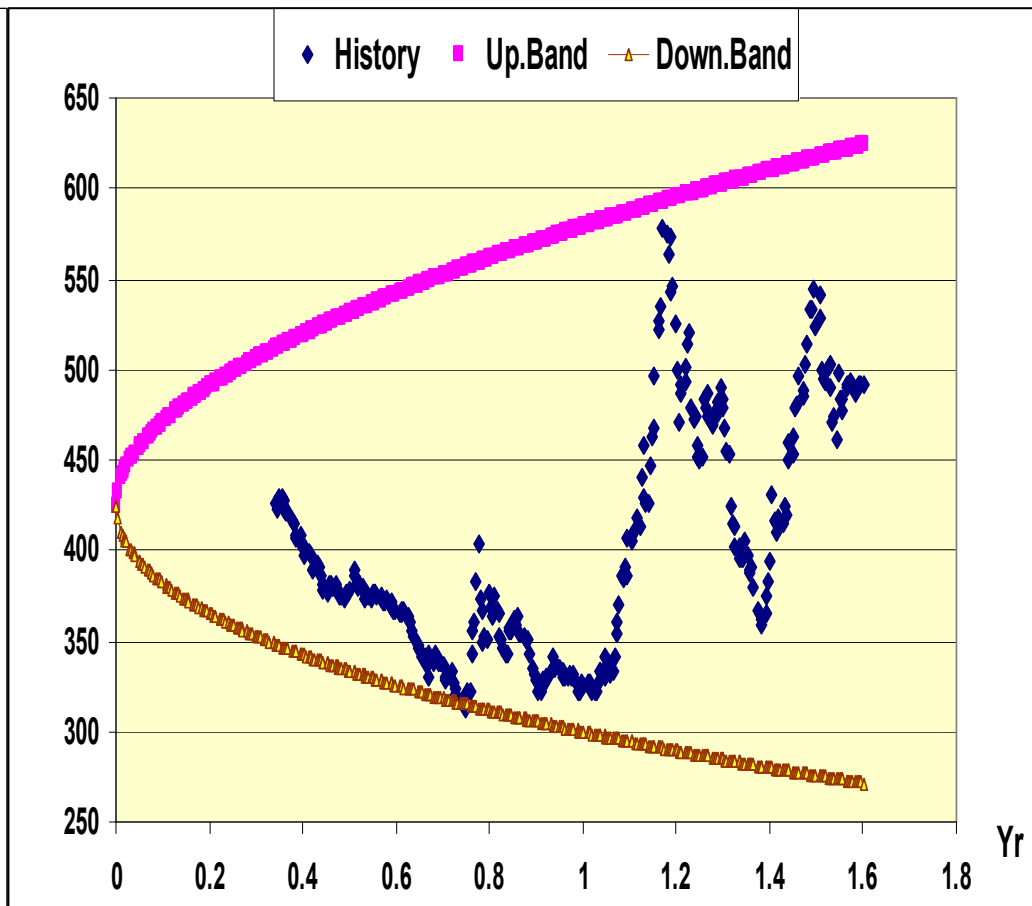
► CDX.IG (2004-2007)

- $\mu = 2\%$
- $Vol = 40\%$



► CDX.HY (2004-2007)

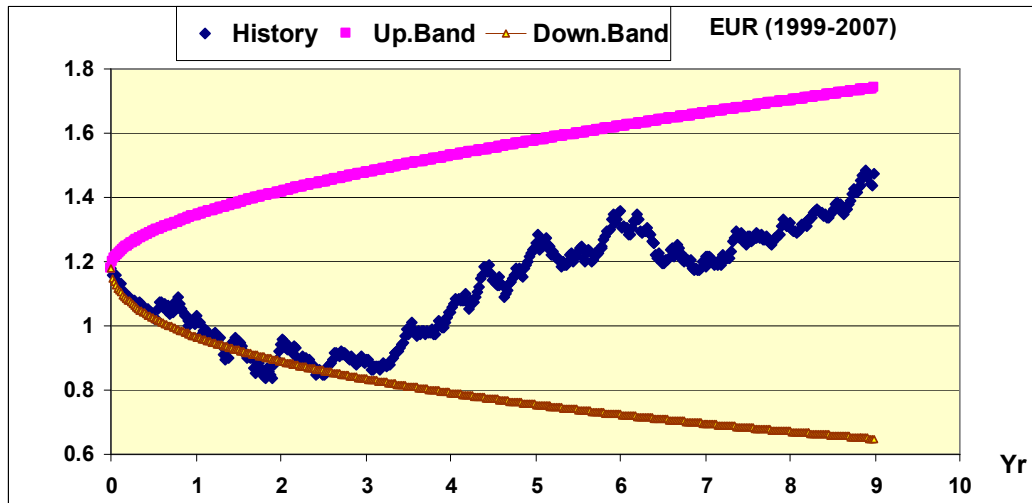
- $\mu = 0\%$
- $Vol = 20\%$



Historical Time Series (FX Model Band)

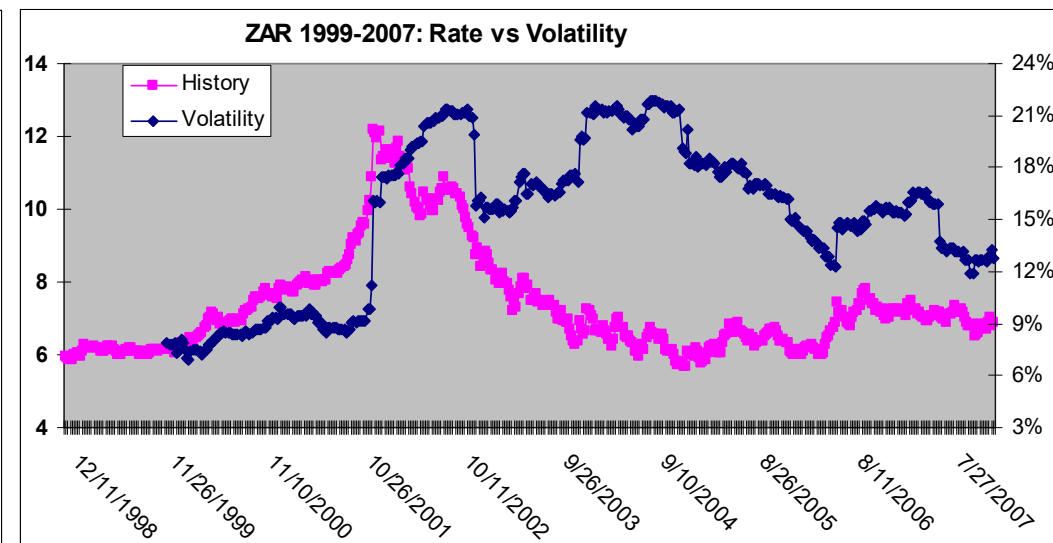
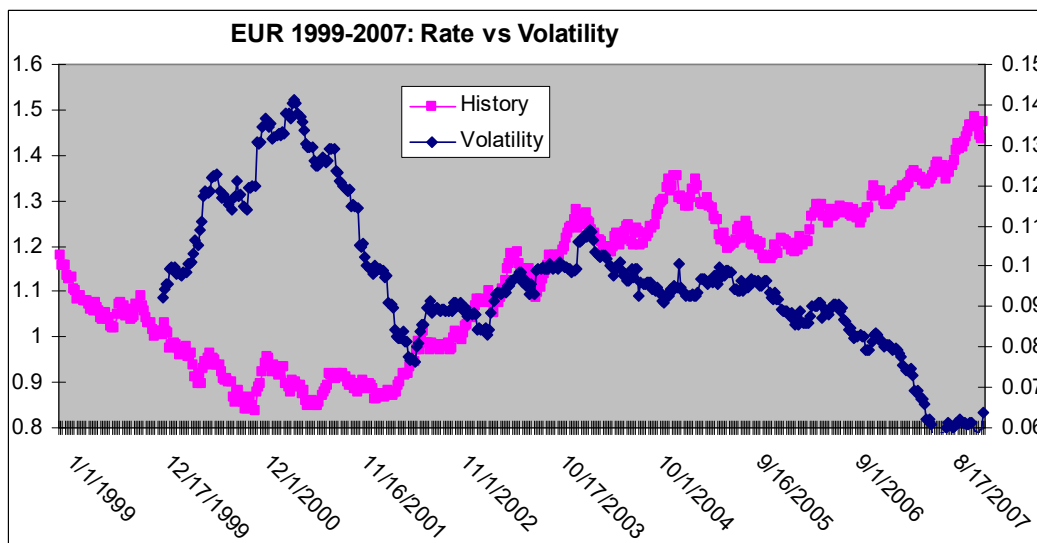
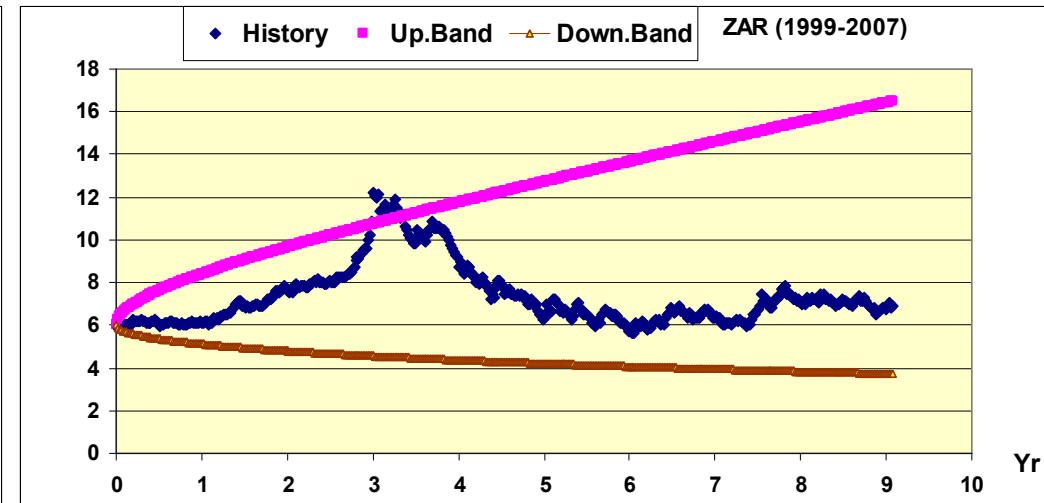
► EUR (G10 Ccy)

— Vol = 7.5%, Drift = -3%



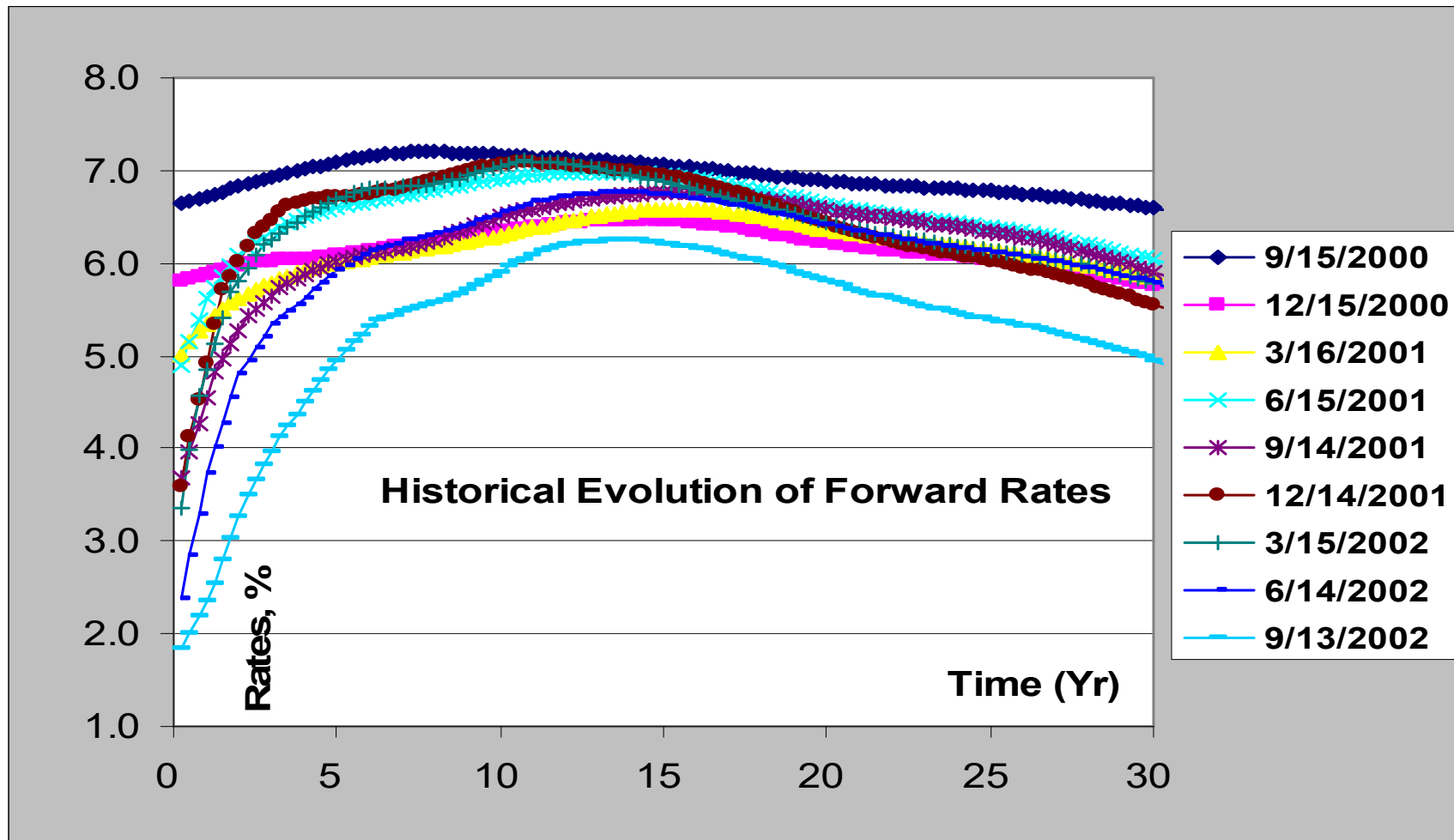
► ZAR (Emkt Ccy)

— Vol = 15%, Drift = 8%



Interest Rate Curves: History

► Historical Fwd Rate Curves: 2000-2002



Single-Factor vs. Multi-factor Models

The Evolution of the Price-Factors can be modeled

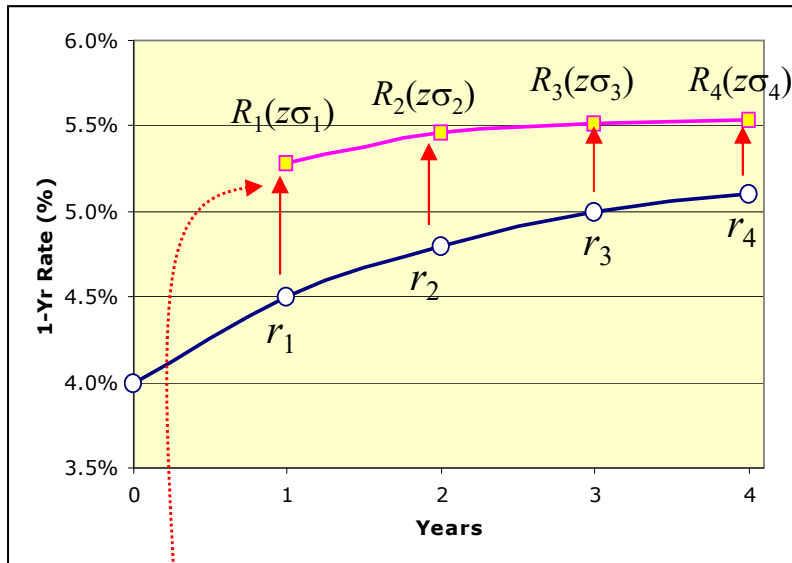
► Single Factor

- Only one risk factor determines the future value of the price factor: 1-Factor Log-normal Model
- Example: FX Spot Rate, Equity Price
- Example: Parallel Shocks on IR Fwd Rate Curve

► Multi-Factor

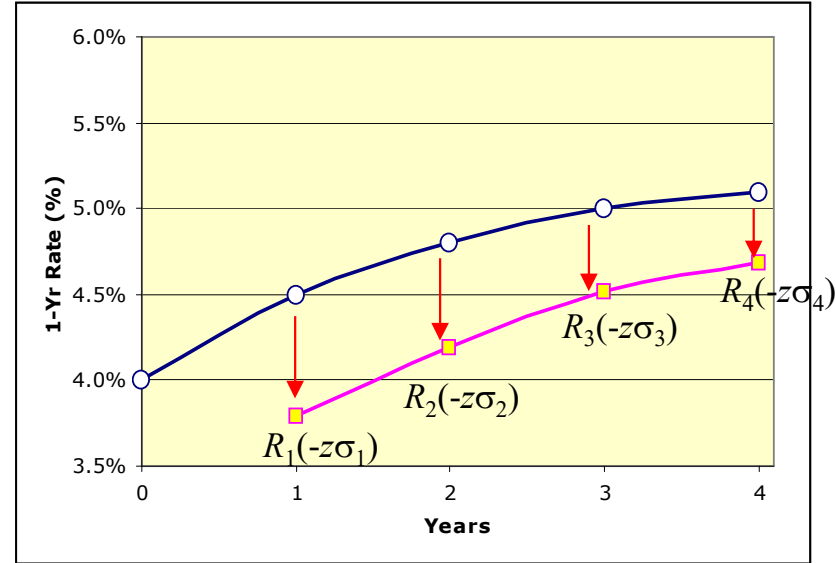
- Two or more risk factors determine the future value of the price factor: 3-Factor PCA Model
- Example: Interest Rate Curve, Crude/NG Price Curve
- Example: Parallel, Twist and Tilt Shapes of Fwd Curve

Single Factor Interest Rate Curve

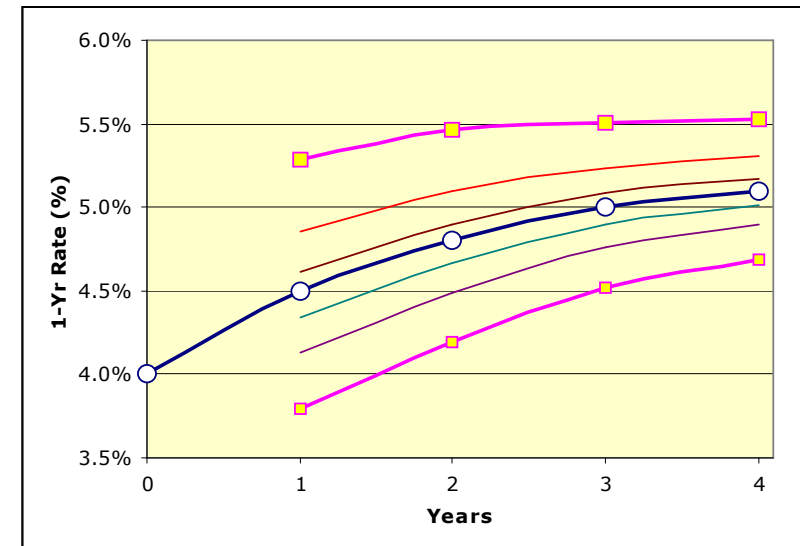


$$R_1(z\sigma_1) = r_1 \exp[-0.5\sigma^2 + z\sigma_1]$$

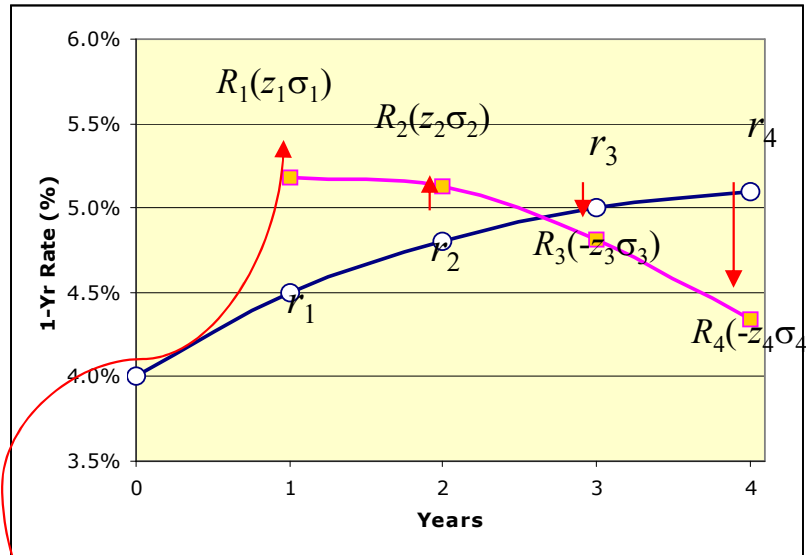
- In single-factor model, we use the same random number z for all parts of the curve.
- The random number z determines the direction of the curve.
- When z is positive, the whole curve shifts up and shifts down when z is negative.



Possible shapes of 1-yr Forward Rates

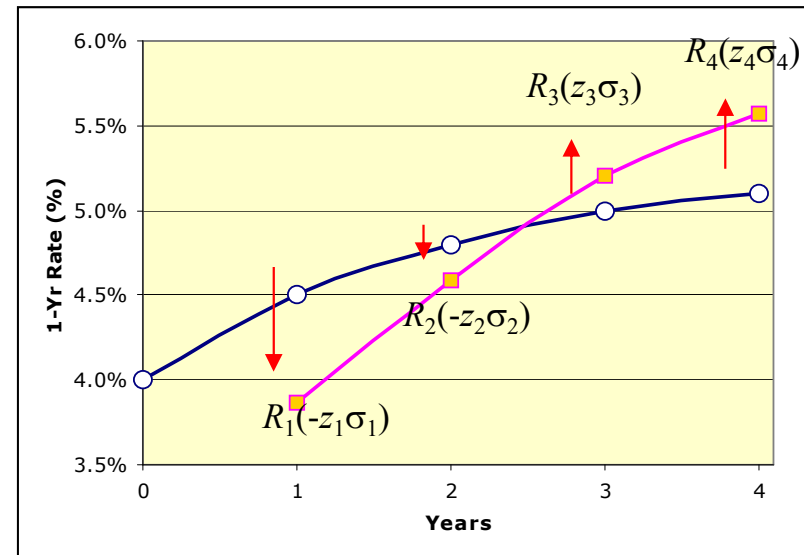


Multi Factor Interest Rate Curve

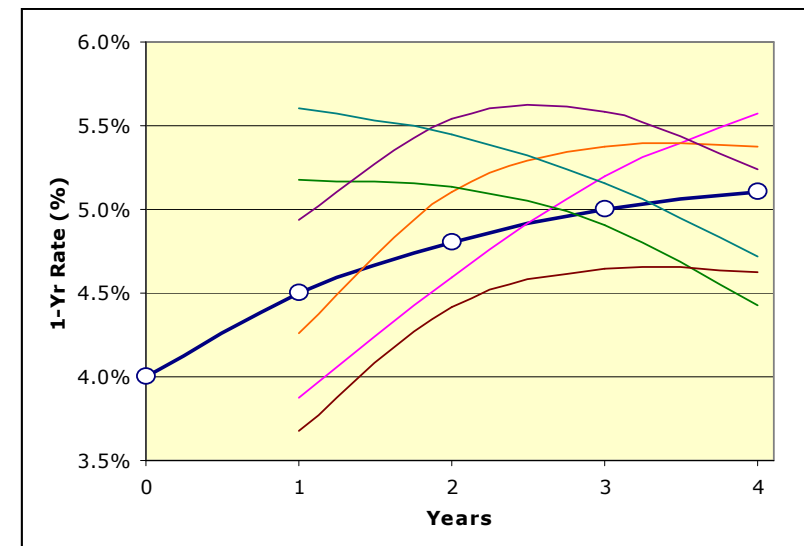


$$R_1(z_1\sigma_1) = r_1 \exp[-0.5\sigma_1^2 + z_1\sigma_1]$$

- The random numbers z_i 's determine the direction of the curve.
- In multi-factor model, we use different z_i 's for each part of the curve.
- When z_i is positive, the particular forward rate shifts up and shifts down when z_i is negative.



Possible shapes of 1-yr Forward Rates



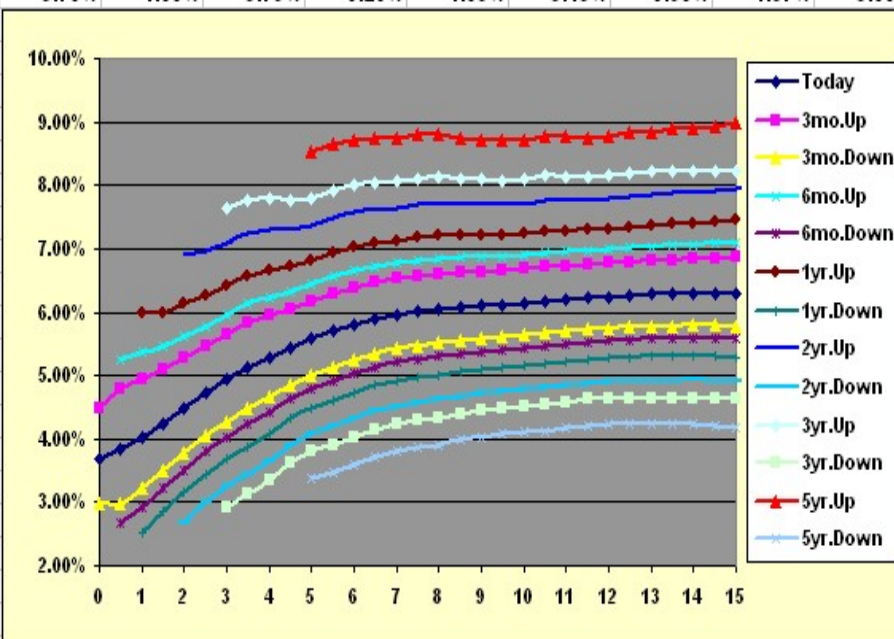
One-Factor Fwd Curve Model

Fwd Curve 1-Factor Scenario Model

$$F_k(t) = F_k(0) \exp \left[-\frac{1}{2} \sigma_k^2 t + z \sigma_k \sqrt{t} \right]$$

w/o Mean-Reversion

Yrs	Forward	0.25	0.25	0.25	0.5	0.5	0.5	1	1	1	2	2	2	3	3	3	5	5	5
Term	Today	3mo.Up	3mo.Mid	3mo.Dow	6mo.Up	6mo.Mid	6mo.Dow	1yr.Up	1yr.Mid	1yr.Dow	2yr.Up	2yr.Mid	2yr.Dow	3yr.Up	3yr.Mid	3yr.Dow	5yr.Up	5yr.Mid	5yr.Dow
0	3.680%	4.48%	3.65%	2.98%															
0.5	3.824%	4.80%	3.78%	2.98%	5.25%	3.74%	2.67%												
1	4.020%	4.94%	3.99%	3.21%	5.36%	3.95%	2.91%	5.98%	3.88%	2.52%									
1.5	4.249%	5.08%	4.22%	3.51%	5.45%	4.20%	3.23%	6.00%	4.14%	2.86%									
2	4.490%	5.28%	4.47%	3.78%	5.63%	4.44%	3.51%	6.14%	4.40%	3.15%	6.91%	4.31%	2.69%						
2.5	4.722%	5.46%	4.70%	4.05%	5.78%	4.68%	3.79%	6.26%	4.65%	3.45%	6.96%	4.57%	3.00%						
3	4.932%	5.64%	4.91%	4.28%										7.63%	4.73%	2.93%			
3.5	5.119%	5.82%	5.10%	4.47%										7.77%	4.93%	3.12%			
4	5.286%	5.95%	5.27%	4.67%										7.80%	5.12%	3.36%			
4.5	5.438%	6.06%	5.43%	4.86%										7.75%	5.29%	3.61%			
5	5.576%	6.17%	5.56%	5.02%										7.80%	5.44%	3.80%	8.52%	5.36%	3.37%
5.5	5.699%	6.30%	5.69%	5.14%										7.93%	5.57%	3.91%	8.65%	5.48%	3.48%
6	5.805%	6.40%	5.79%	5.25%										8.00%	5.68%	4.04%	8.71%	5.60%	3.60%
6.5	5.892%	6.47%	5.88%	5.35%										8.04%	5.77%	4.15%	8.73%	5.70%	3.72%
7	5.959%	6.53%	5.95%	5.42%										8.07%	5.85%	4.24%	8.75%	5.77%	3.81%
7.5	6.009%	6.58%	6.00%	5.47%										8.11%	5.90%	4.29%	8.79%	5.82%	3.86%
8	6.046%	6.61%	6.04%	5.51%										8.12%	5.94%	4.34%	8.79%	5.87%	3.91%
8.5	6.073%	6.62%	6.06%	5.55%										8.10%	5.97%	4.40%	8.75%	5.90%	3.98%
9	6.097%	6.64%	6.09%	5.58%										8.09%	6.00%	4.45%	8.73%	5.93%	4.03%
9.5	6.122%	6.65%	6.11%	5.62%										8.08%	6.03%	4.49%	8.71%	5.96%	4.08%
10	6.148%	6.68%	6.14%	5.65%										8.09%	6.05%	4.53%	8.72%	5.99%	4.12%
10.5	6.174%	6.71%	6.17%	5.66%										8.15%	6.08%	4.53%	8.78%	6.01%	4.12%
11	6.200%	6.73%	6.19%	5.70%										8.14%	6.11%	4.58%	8.76%	6.04%	4.17%
11.5	6.224%	6.74%	6.22%	5.73%										8.13%	6.13%	4.63%	8.74%	6.07%	4.22%
12	6.245%	6.77%	6.24%	5.75%										8.16%	6.15%	4.64%	8.77%	6.09%	4.23%
12.5	6.262%	6.79%	6.25%	5.76%	7.02%	6.25%	5.56%	7.34%	6.23%	5.29%	7.82%	6.20%	4.91%	8.20%	6.17%	4.64%	8.82%	6.11%	4.23%
13	6.277%	6.81%	6.27%	5.77%	7.03%	6.26%	5.57%	7.36%	6.25%	5.30%	7.84%	6.21%	4.92%	8.22%	6.18%	4.65%	8.84%	6.12%	4.24%
13.5	6.288%	6.82%	6.28%	5.78%	7.05%	6.27%	5.58%	7.39%	6.26%	5.30%	7.87%	6.22%	4.92%	8.22%	6.18%	4.65%	8.89%	6.13%	4.23%
14	6.297%	6.83%	6.29%	5.79%	7.06%	6.28%	5.58%	7.40%	6.26%	5.31%	7.88%	6.23%	4.93%	8.22%	6.18%	4.65%	8.90%	6.14%	4.23%
14.5	6.302%	6.84%	6.29%	5.79%	7.08%	6.29%	5.58%	7.41%	6.27%	5.30%	7.91%	6.24%	4.92%	8.22%	6.18%	4.65%	8.94%	6.14%	4.22%
15	6.305%	6.86%	6.30%	5.78%	7.10%	6.29%	5.57%	7.44%	6.27%	5.28%	7.94%	6.24%	4.90%	8.22%	6.18%	4.65%	9.00%	6.14%	4.19%



Simulating IR Curves Using PCA

- ▶ Historical weekly fwd rate changes are decomposed into PCs:
 - $\log(F(n+1)/F(n)) = \mu_1(n) \cdot PC_1 + \mu_2(n) \cdot PC_2 + \mu_3(n) \cdot PC_3$
- ▶ Find weekly volatilities from obtained series $\mu_1(n)$, $\mu_2(n)$ and $\mu_3(n)$. Scale them to obtain yearly volatilities σ_1 , σ_2 and σ_3 .
- ▶ Model forward rate curve evolution using PCs and their Vols:
 - $F(t; t_1, t_2) = F(0; t_1, t_2) \cdot \exp(\sigma_1 \cdot w_1(t) \cdot PC_1(t_1, t_2) +$
 - $\sigma_2 \cdot w_2(t) \cdot PC_2(t_1, t_2) +$
 - $\sigma_3 \cdot w_3(t) \cdot PC_3(t_1, t_2) - 0.5 \cdot t \cdot \text{drift})$
- ▶ Note PCs are uncorrelated by the virtue of the PCA process!

A simulation model based on PCA framework is more efficient than other multi-factor approaches, and it generates more realistic path evolution than the single-factor models that is paramount in our application of the exposure calculation on derivative transactions.

Simulating Interest Rate Curve Using PCA

1. Perform PCA to find “Eigenvalues” and “Eigenvectors” of the covariance matrix.
2. Determine the number of relevant PCA factors (usually no more than three factors).
3. Draw uncorrelated random variates from a standard normal distribution for each component.
4. Scale each random variate from its associated factor score and by the square root of the time step.
5. Load the scaled value step onto each forward rates and prices using the “factor loadings”. The return for each forward rates and prices will then be given by the sum of the product of each scales value in previous step multiplied by the “factor loading”.

A simulation model based on PCA framework is more efficient than other multi-factor approaches, and it generates more realistic path evolution than the single-factor models that is paramount in our application of the exposure calculation on derivative transactions.

Three-Factor PCA Fwd Curve Model

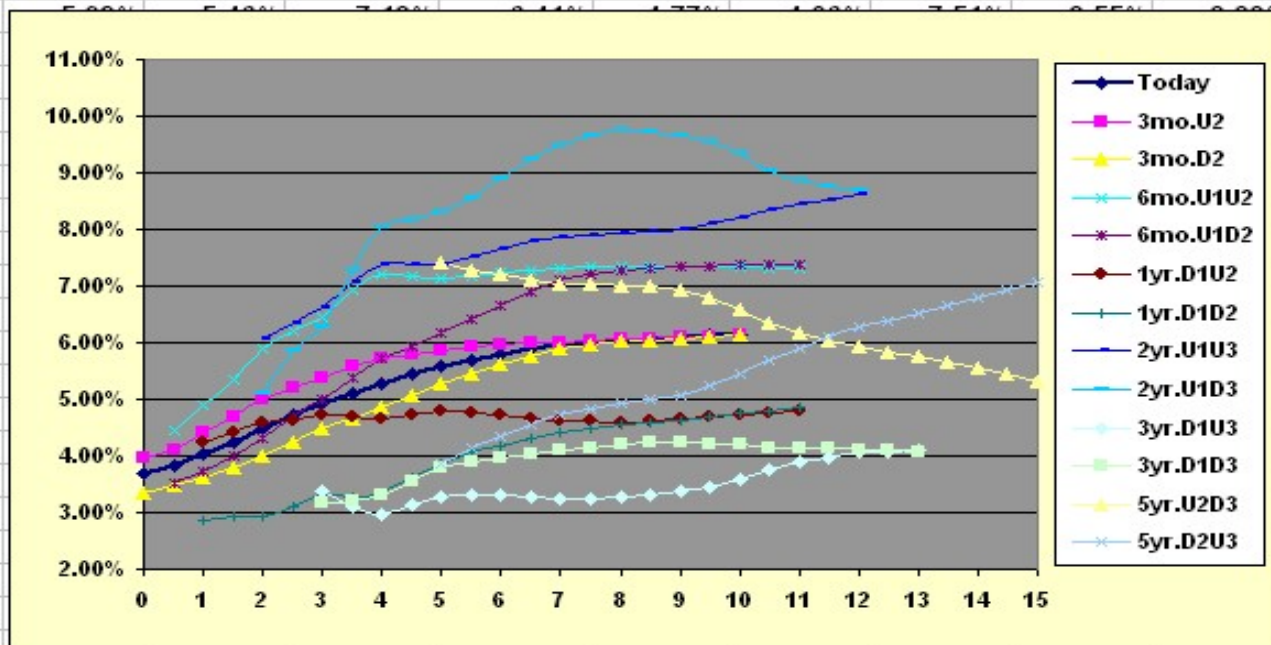
Fwd Curve 3-Factor PCA Scenario Model

MeanReversion 10%

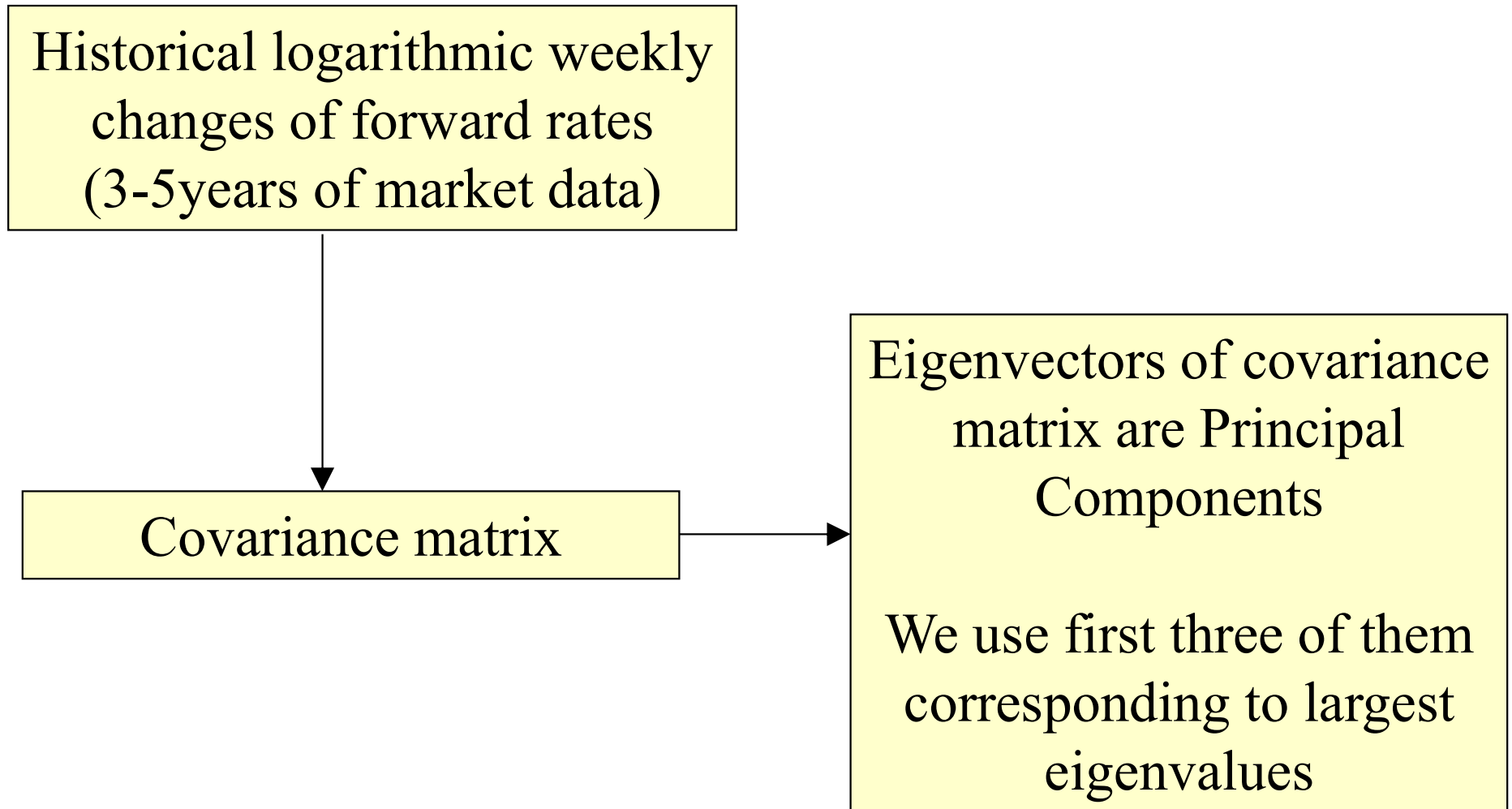
w/o Mean-Reversion

$$F_k(t) = F_k(0) \exp \left(\sum_{i=1}^3 z^i(t) \cdot PC_k^i \cdot \alpha^i - \frac{1}{2} t \sum_{i=1}^3 (PC_k^i \cdot \alpha^i)^2 \right)$$

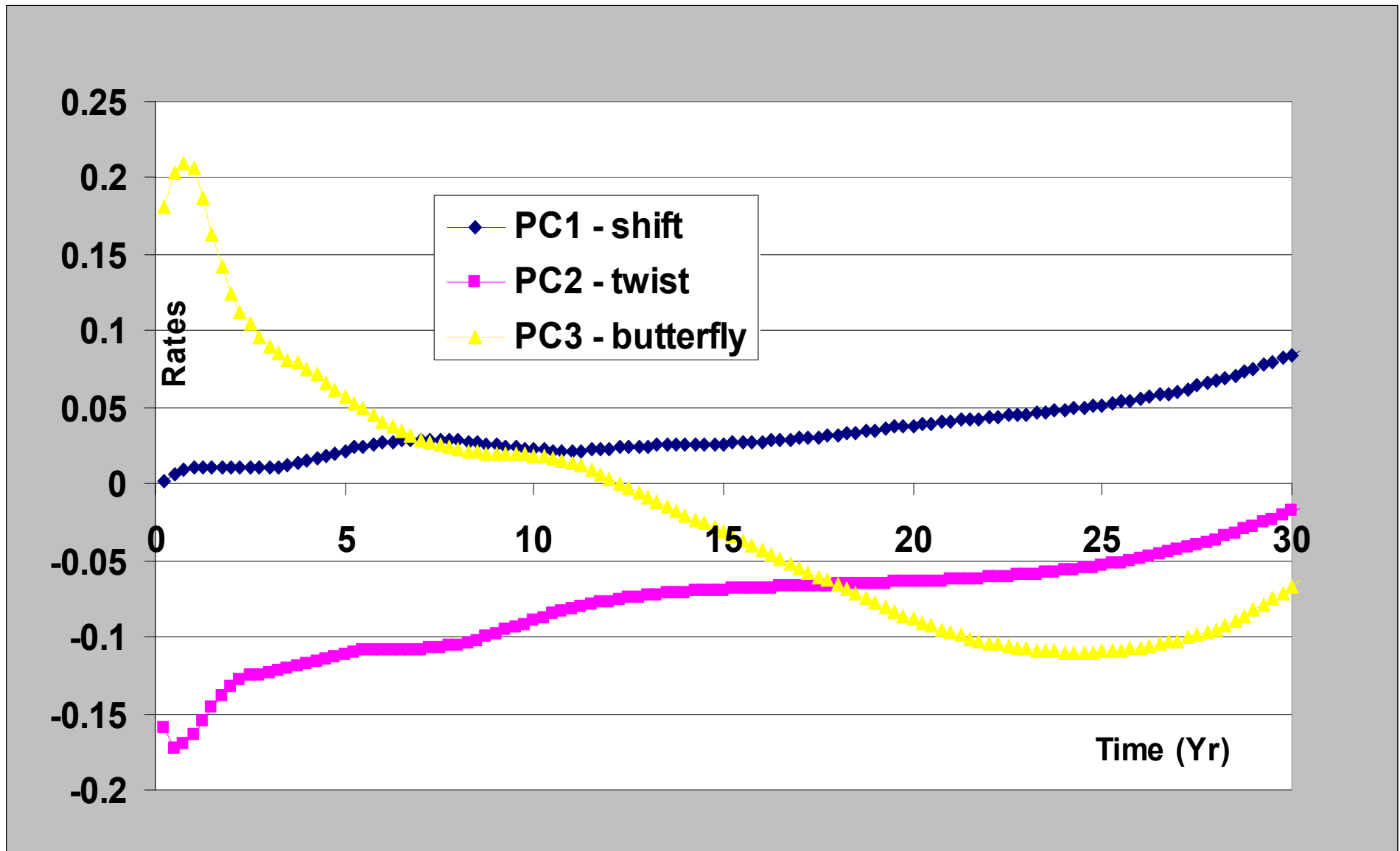
Yrs		0.25	0.25	0.5	0.5	1	1	2	2	3	3	5	5
Term	Today	3mo.U2	3mo.D2	6mo.U1U2	6mo.U1D2	1yr.D1U2	1yr.D1D2	2yr.U1U3	2yr.U1D3	3yr.D1U3	3yr.D1D3	5yr.U2D3	5yr.D2U3
0	3.680%	3.97%	3.36%										
0.5	3.824%	4.11%	3.48%	4.44%	3.51%								
1	4.020%	4.40%	3.61%	4.91%	3.71%	4.24%	2.86%						
1.5	4.249%	4.68%	3.81%	5.36%	4.00%	4.43%	2.93%						
2	4.490%	4.99%	4.00%	5.91%	4.31%	4.59%	2.94%	6.06%	5.09%				
2.5	4.722%	5.19%	4.26%	6.22%	4.70%	4.63%	3.11%	6.34%	5.85%				
3	4.932%	5.38%	4.49%	6.45%	5.00%	4.72%	3.30%	6.61%	6.29%	3.39%	3.19%		
3.5	5.119%	5.58%	4.67%	6.92%	5.37%	4.67%	3.27%	7.08%	7.26%	3.11%	3.21%		
4	5.286%	5.72%	4.86%	7.21%	5.71%	4.65%	3.35%	7.39%	8.03%	2.98%	3.30%		
4.5	5.438%	5.80%	5.08%	7.17%	5.94%	4.73%	3.62%	7.39%	8.16%	3.15%	3.56%		
5	5.576%	5.86%	5.28%	7.14%	6.16%	4.78%	3.88%	7.38%	8.30%	3.28%	3.79%	7.43%	3.87%
5.5	5.699%											3.90%	7.28%
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9.5	6.122%											4.22%	6.78%
10	6.148%											4.22%	6.59%
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11	6.200%											4.14%	6.35%
11.5	6.224%											4.13%	6.18%
12	6.245%											4.12%	6.04%
12.5	6.262%											4.11%	5.92%
13	6.277%											4.10%	5.84%
13.5	6.288%											4.08%	5.76%
14	6.297%												5.66%
14.5	6.302%												5.55%
15	6.305%												5.32%



Principal Component Analysis



Shape of PCA shock factors

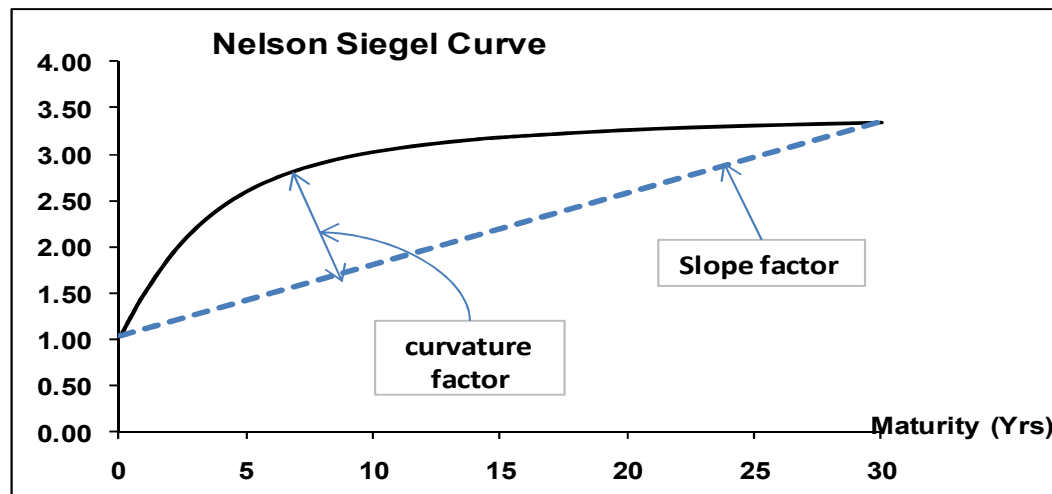


Interest Rate Model – Parametric Approach

Nelson Siegel Model (1987)

$$y(\tau) = b_0 + b_1 \times \frac{1 - e^{-\lambda\tau}}{\lambda\tau} + b_2 \times \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right)$$

- A yield curve, the target curve, is associated with one or more interest rate curve(s), the explanatory curve(s), which is(are) provided by the EST Team
- For the target curve, parameters b_0 , b_1 and b_2 indicate the *level*, *slope* and *curvature*



- τ represents monthly time steps, and common practice is to set $\lambda = 0.0609$
- Parameters b_0 , b_1 and b_2 are estimated by quantile regression on monthly historical data (with 10 years of history)

Charles Nelson & Andrew Siegel (1987), Parsimonious Modeling of Yield Curves, Journal Of Business, Vol.60, No.4, oo473-489.